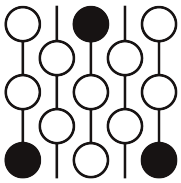
Review of Problem 2 – Astra

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

The paper gives the correct solution of the requested minimax rate (up to log factors) using novel ideas for estimation and lower bound alternatives, compared to the published literature. The estimation relies on a plugin approach for the explicit reconstruction and detailed arguments to derive the correct convergence rate. The error decomposition is different from the human proof and simplifies some of their upper bounds. The lower bound relies on spherical perturbations in the image domain of the underlying nonlinear forward operator.

Correctness. The announced solution of the problem is correct. The decomposition into separate proof steps is very reasonable and well done. The more technical steps are sometimes too quickly executed to be checked without much further efforts, especially for the upper bound. Yet, they seem flawless and are a genuine, innovative contribution to the field.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition. Compared to the correctness part this point is much weaker. Several times details for the proofs are missing and tools used are not further explained. The structure of the results is suboptimal: for instance, the estimator and the main error decomposition are hidden within a lengthy proof. They come after the main technical lemmata and bounds, which are thus not motivated and lack intuition. There are no further explanations given along the proofs and only the final Remark 8.3 delivers some intuition. The notation is ok, not perfect, but you can follow when checking definitions in prior parts. An overview section is missing, but would very valuable.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions. Only four references are cited. There is no reference to literature on prior (suboptimal) solutions to the problem. Several tools from analysis and probability are used without giving reference and explaining the inherent notation (e.g. real interpolation and K-functionals).

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

Abstract

For a sufficiently small, fixed Hölder–Zygmund ball of conductivities around the constant function 1, the minimax squared L^2-risk is, up to logarithmic factors,

$$n^{-2\alpha/(2\alpha+d+3)}.$$

We prove matching polynomial upper and lower bounds for every $d \geq 2$ and $\alpha > 1$. The upper bound uses a regularized reconstruction along gradient trajectories, together with a variance estimate that exploits integration along these trajectories. The lower bound uses perturbations that oscillate in the $d-1$ angular directions. No boundary condition on the conductivity is imposed.

3 Statement and conventions

Let $D = B(0,1) \subset \mathbb{R}^d$. We interpret $C^s(D)$ as the Hölder–Zygmund restriction space $B^s_{\infty,\infty}(D)$. Any fixed usual equivalent norm may be used; the constants below, including the admissible radius, may depend on this choice.

White noise means the isonormal experiment

$$Y_n(\varphi) = \langle u_f, \varphi \rangle_{L^2(D)} + n^{-1/2}\mathbb{W}(\varphi), \qquad \varphi \in L^2(D),$$

where $\mathbb{E}[\mathbb{W}(\varphi)\mathbb{W}(\psi)] = \langle \varphi, \psi \rangle$. Write

$$\mathcal{F}(\alpha, r) = \{f : \|f - \mathbf{1}\|_{C^\alpha(D)} \leq r\}, \qquad R_n(\alpha, r) = \inf_{\widehat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\widehat{f} - f\|^2_{L^2(D)}.$$

Theorem 3.1. *Fix $d \geq 2$ and $\alpha > 1$. There exists $r_* > 0$ such that, for each fixed $r \in (0, r_*]$, there are constants $c, C > 0$ and n_0 for which*

$$c\, n^{-2\alpha/(2\alpha+d+3)} \leq R_n(\alpha, r) \leq C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)}, \qquad n \geq n_0.$$

If $\alpha \notin \mathbb{N}$, the logarithmic factor in this upper bound can be omitted. In particular, the answer up to logarithmic factors is

$$\boxed{R_n(\alpha, r) = \eqsim n^{-2\alpha/(2\alpha+d+3)}.}$$

> **Reviewer 1**
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> Symbol in boxed formula weird and not introduced.

For a smooth function a, the notation

$$\|a\|_{k,\infty;\Omega} = \max_{|\beta| \leq k} \|\partial^\beta a\|_{L^\infty(\Omega)}$$

denotes the classical derivative norm, not an integer-index Zygmund norm. Domain subscripts will be suppressed when unambiguous. Constants denoted by C may change from line to line, but are independent of n, the bandwidth, and the conductivity in the specified parameter class.

4 Analytic preliminaries

4.1 Function spaces and the forward equation

We use the Littlewood–Paley characterization

$$\|a\|_{C^s(\mathbb{R}^d)} \asymp \sup_{j\ge 0} 2^{js}\|\Delta_j a\|_\infty \tag{1}$$

for an inhomogeneous smooth dyadic decomposition. The equivalence with classical Hölder spaces at noninteger positive indices, and the differentiation properties of these spaces, are given in [3, Chapter 13, Section 8].

We also use a common bounded extension operator from a smooth bounded domain to $\mathbb{R}^d$, bounded on all the finitely many Besov spaces under consideration. Such an operator exists even for Lipschitz domains; see [2].

Here are some consequences, with the relevant arguments recalled. Differentiation maps C^s boundedly to C^{s-1}. For $s > 0$, C^s is a multiplication algebra:

$$\|ab\|_{C^s} \le C_s\|a\|_{C^s}\|b\|_{C^s}. \tag{2}$$

Indeed, decompose the product into low–high, high–low, and comparable-frequency products. The low–high contribution to its j-th dyadic block is bounded by $C\|a\|_\infty 2^{-js}\|b\|_{C^s}$; interchange a, b for the high–low contribution. For comparable frequencies $k \ge j - C$, the bound is

$$C\|a\|_{C^s}\|b\|_{C^s}\sum_{k\ge j-C} 2^{-2ks} \le C_s 2^{-js}\|a\|_{C^s}\|b\|_{C^s}.$$

This proves (2). Extension and restriction give the same facts on D.

We shall also use the real interpolation identity

> **Reviewer 1**
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> Reference for real interpolation and thus notation used!

$$(C^{s_0}, C^{s_1})_{\theta,\infty} = C^{(1-\theta)s_0+\theta s_1}, \qquad s_0 < s_1, \quad 0 < \theta < 1. \tag{3}$$

For completeness, on $\mathbb{R}^d$ this follows by splitting the dyadic series at a level J. If $s = (1-\theta)s_0 + \theta s_1$, the low and high parts satisfy

$$\|a_{\le J}\|_{C^{s_1}} \le C2^{J(s_1-s)}\|a\|_{C^s}, \qquad \|a_{>J}\|_{C^{s_0}} \le C2^{-J(s-s_0)}\|a\|_{C^s}.$$

Taking $2^{J(s_1-s_0)} \asymp t^{-1}$ bounds the K-functional by $Ct^\theta\|a\|_{C^s}$. Conversely, applying Δ_j to any decomposition $a = a_0 + a_1$ gives

$$\|\Delta_j a\|_\infty \le C2^{-js_0}K(2^{-j(s_1-s_0)}, a; C^{s_0}, C^{s_1}),$$

which proves the reverse bound. A common extension operator transfers the identity to D.

Lemma 4.1 (Uniform perturbative regularity)**.** *Put*

$$u_\circ(x) = \frac{|x|^2 - 1}{2d}.$$

For sufficiently small $r_* > 0$, every $f \in \mathcal{F}(\alpha, r_*)$ satisfies $1/2 \le f \le 3/2$, and the Dirichlet problem

$$\operatorname{div}(f\nabla u) = 1, \qquad u|_{\partial D} = 0$$

has a unique weak solution. Moreover,

$$\|u_f - u_\circ\|_{C^{\alpha+1}(D)} \le C\|f - \mathbf{1}\|_{C^\alpha(D)}. \tag{4}$$

> **Reviewer 1**
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> This lemma should be cited from references or – if not exactly stated as is – compared to literature.

Proof. Let $\mathcal{P}F$ be the zero-Dirichlet solution of $\Delta w = \operatorname{div} F$. For every $s > 1$,

$$\|\mathcal{P}F\|_{C^{s+1}(D)} \le C_s \|F\|_{C^s(D)}. \tag{5}$$

At noninteger s, this is the classical global Schauder estimate and solvability theorem, including higher-order regularity; see [1, Theorems 6.6, 6.14 and 6.19]. At integer s, apply these bounds at two surrounding noninteger indices and use (3). Thus (5) holds also at the Zygmund indices.

Writing $u = u_\circ + w$, the desired equation becomes

$$w = -\mathcal{P}\big((f-1)\nabla u_\circ\big) - \mathcal{P}\big((f-1)\nabla w\big).$$

By (2) and (5), the second term is a linear operator on the zero-trace subspace of $C^{\alpha+1}(D)$ with norm at most $C\|f-1\|_{C^\alpha}$. Choosing r_* so that this norm is at most $1/2$, the Neumann series gives a solution and (4).

The embedding $C^\alpha(D) \hookrightarrow C^0(\overline{D})$ allows us to decrease r_* so that $1/2 \le f \le 3/2$. Uniqueness of a weak solution follows by testing the equation for the difference of two solutions against that difference: $\int_D f|\nabla w|^2 = 0$. Finally, the constructed solution has zero trace and belongs to $H^1_0(D)$, so it is that unique weak solution. □

4.2 Smoothing and a product cancellation

Set

$$m_0 = \lceil \alpha \rceil - 1.$$

Fix a real $K \in C^\infty_c(B(0,1))$ such that

$$\int K = 1, \qquad \int x^\beta K(x)\,dx = 0 \quad (1 \le |\beta| \le 2m_0 + 2). \tag{6}$$

Such a kernel can be obtained from a finite linear combination of distinct dilations of a smooth unit-integral bump: the required coefficients solve a Vandermonde system in the dilation parameters. Positivity of K is neither required nor asserted.

For $0 < h < 1/8$, put

$$K_h(x) = h^{-d}K(x/h), \qquad \Omega_h = B(0, 1-2h), \qquad S_h a(x) = \int_D K_h(x-y)a(y)\,dy, \quad x \in \Omega_h,$$

and let $L_h = \log(e/h)$.

Lemma 4.2 (Smoothing and commutator bounds)**.** *Suppose $a, b \in C^\alpha(D)$ have bounded norms. Uniformly on Ω_h,*

$$\|S_h a - a\|_\infty \le C h^\alpha L_h, \tag{7}$$

$$\big\|\partial^\gamma\big((S_h a)(S_h b) - S_h(ab)\big)\big\|_\infty \le C h^{\alpha+1-|\gamma|} L_h^2, \qquad |\gamma| \le 2. \tag{8}$$

If $\alpha \notin \mathbb{N}$, all factors L_h in these two bounds may be replaced by 1. The statements hold componentwise for vector-valued b.

Proof. We first record the local Taylor remainder

$$\sup_{|y-x|\le h} |a(y) - P_{a,x}(y)| \le Ch^{\alpha} L_h, \tag{9}$$

where $P_{a,x}$ is the Taylor polynomial of degree m_0. Its coefficients are uniformly bounded. At noninteger α, this is the ordinary Hölder Taylor estimate, without L_h. At integer α, derivatives of order $\alpha - 1$ have modulus of continuity bounded by $Ct\log(e/t)$. To see the latter directly from (1), split their dyadic series at $2^J \asymp t^{-1}$: the low-frequency increments sum to $C\sum_{j\le J} t$, and the high-frequency tail is at most $C\sum_{j>J} 2^{-j}$. Taylor's integral remainder then gives (9). Extension to $\mathbb{R}^d$ justifies this argument uniformly near the boundary.

Polynomial reproduction from (6) immediately proves (7). To prove (8), write

$$Q_h(a,b) = (S_h a)(S_h b) - S_h(ab).$$

Fix x and write $a = P_a + R_a$, $b = P_b + R_b$, using the Taylor polynomials at x. Keep these polynomials fixed when differentiating at x. Polynomial reproduction gives $Q_h(P_a, P_b) = 0$. Also $Q_h(a, c) = 0$ for every constant c.

On $B(x,h)$, the remainders satisfy $|R_a| + |R_b| \le Ch^{\alpha} L_h$, whereas $|P_a - P_a(x)| + |P_b - P_b(x)| \le Ch$. Differentiating the convolution kernels gives, for $|\gamma| \le 2$,

$$|\partial^{\gamma} Q_h(R_a, P_b)(x)| + |\partial^{\gamma} Q_h(P_a, R_b)(x)| \le Ch^{\alpha+1-|\gamma|} L_h.$$

For example, in the first expression subtract the constant $P_b(x)$; the differentiated convolution of $R_a(P_b - P_b(x))$ has precisely this bound. In the product term, a derivative falling on the polynomial is bounded by Ch^{1-k} at derivative order $k \ge 1$; the undifferentiated polynomial vanishes at x. Finally,

$$|\partial^{\gamma} Q_h(R_a, R_b)(x)| \le Ch^{2\alpha-|\gamma|} L_h^2.$$

Since $\alpha > 1$, these inequalities prove (8). The same proof without logarithms applies at noninteger α. □

Lemma 4.3 (Completion of a thin boundary strip)**.** *For all sufficiently small h, there is a linear map $E_h : L^2(\Omega_h) \to L^2(D)$ such that*

$$\|E_h a\|_{L^2(D)} \le C\|a\|_{L^2(\Omega_h)}, \tag{10}$$

$$\|E_h(f|_{\Omega_h}) - f\|_{L^2(D)} \le C\|f\|_{C^{\alpha}(D)} h^{\alpha} L_h. \tag{11}$$

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> *Since the left-hand side of (11) is in L^2, we might win a factor $h^{1/2}$: for instance for $\alpha = 0$ use $|D \setminus \Omega_h| \lesssim h$ and $E_h f$ remains bounded.*

At noninteger α, L_h can be omitted.

Proof. Let $q = m_0 + 1$ and $\lambda_j = (-1)^{j+1}\binom{q}{j}$, $1 \le j \le q$. Then

$$\sum_{j=1}^{q} \lambda_j = 1, \qquad \sum_{j=1}^{q} \lambda_j j^k = 0, \quad 1 \le k \le m_0.$$

Define $E_h a = a$ on Ω_h, and, for $x \in D \setminus \Omega_h$, define

$$E_h a(x) = \sum_{j=1}^{q} \lambda_j a((1 - 3jh)x).$$

For $h < 1/(6q)$, all arguments on the right belong to Ω_h and all dilation factors are at least $1/2$. A change of variables proves (10). Taylor expansion of f at x, together with the displayed moment identities and (9), proves (11).

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> This proof is just a sketch, but not rigorous.

□

5 Inversion along gradient trajectories

For a smooth function v, define

$$T_v a = \operatorname{div}(a\nabla v) = \nabla v \cdot \nabla a + (\Delta v)a.$$

Lemma 5.1 (Transport inverse)**.** *Let $\Omega = B(0,R)$, $3/4 \le R \le 1$, and suppose v is smooth on a neighborhood of $\overline{\Omega}$, with*

$$\lambda I \le D^2 v \le \Lambda I, \qquad \nabla v \cdot \nu > 0 \text{ on } \partial\Omega, \tag{12}$$

where $0 < \lambda \le \Lambda$. There is a unique critical point $p \in \Omega$.

> ***Reviewer 1***
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> *Specify 'critical point of v'. The result is known in the literature: cite!*

Let $z_t(x)$ solve

$$\frac{d}{dt} z_t(x) = -\nabla v(z_t(x)), \qquad z_0(x) = x.$$

This flow exists in $\overline{\Omega}$ for all $t \ge 0$, and

$$e^{-\Lambda t}|x-p| \le |z_t(x) - p| \le e^{-\lambda t}|x-p|. \tag{13}$$

For smooth g, the unique bounded smooth solution of $T_v a = g$ is

$$P_v g(x) = \int_0^\infty \exp\left\{ -\int_0^t \Delta v(z_s(x))\, ds \right\} g(z_t(x))\, dt. \tag{14}$$

With $c_0 = d\lambda$, it satisfies

$$\|P_v g\|_\infty \le c_0^{-1} \|g\|_\infty, \tag{15}$$

$$\|P_v g\|_2 \le 2c_0^{-1} \|g\|_2, \tag{16}$$

$$\|P_v g\|_{H^1} \le C\big\{ \|g\|_{H^1} + (1 + \|D^3 v\|_\infty)\|g\|_2 \big\}. \tag{17}$$

The constants in the last estimate depend only on d, λ, Λ.

Proof. Strict convexity gives at most one critical point. A minimum on $\overline{\Omega}$ cannot occur at the boundary, because the inward direction decreases v, so this critical point exists in Ω. The vector field $-\nabla v$ points inward at the boundary; the usual first-exit argument and local existence for a smooth vector field therefore give a global backward-gradient flow in $\overline{\Omega}$.

For $z \in \Omega$,

$$\lambda |z-p|^2 \le (z-p) \cdot \nabla v(z) \le \Lambda |z-p|^2.$$

Applying this to the derivative of $|z_t - p|$ proves (13).

Since $\Delta v \geq c_0$, the integral in (14) converges absolutely and gives (15). It is smooth up to the boundary. Here is a justification of differentiating the infinite-time integral. The first variational equation gives $\|D_x z_t\| \leq e^{-\lambda t}$. Inductively, the equation for $D_x^k z_t$ has homogeneous part $-D^2 v(z_t) D_x^k z_t$, while each forcing term is a product of at least two lower-order flow derivatives. Variation of constants therefore gives $\|D_x^k z_t\| \leq C_k e^{-\lambda t}$. Derivatives of $\int_0^t \Delta v(z_s)\, ds$ of every positive order are consequently bounded uniformly in t. Every differentiated integrand is thus bounded by $C_k e^{-c_0 t}$.

The flow semigroup property, or differentiation after splitting the integral at a finite time, gives $T_v P_v g = g$. If $T_v a = 0$, differentiation along the flow shows

$$a(x) = \exp\left\{ -\int_0^t \Delta v(z_s(x))\, ds \right\} a(z_t(x)).$$

Letting $t \to \infty$ proves uniqueness among bounded solutions.

For $a = P_v g$, integration by parts gives

$$\int_\Omega a T_v a = \frac{1}{2} \int_{\partial\Omega} (\nabla v \cdot \nu) a^2 + \frac{1}{2} \int_\Omega (\Delta v) a^2 \geq \frac{c_0}{2} \|a\|_2^2.$$

This proves (16). Differentiating the equation gives

$$T_v(\nabla a) + D^2 v\, \nabla a = \nabla g - a \nabla \Delta v.$$

The same integration by parts, now for a vector field, yields

$$(c_0/2 + \lambda) \|\nabla a\|_2 \leq \|\nabla g\|_2 + \|\nabla \Delta v\|_\infty \|a\|_2.$$

Together with (16), this proves (17). □

6 Gaussian smoothing and the decisive variance estimate

Define

$$Z_h(x) = n^{-1/2} \mathbb{W}(K_h(x - \cdot)), \qquad x \in \mathbb{R}^d,$$

where the white-noise argument is restricted to D.

Lemma 6.1 (Gaussian smoothing bounds)**.** *For every fixed nonnegative integer k and $p \geq 1$,*

$$\mathbb{E}\|Z_h\|_{k,\infty;\Omega_h}^p \leq C_p n^{-p/2} h^{-p(d/2+k)} L_h^{p/2}, \tag{18}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^2 \leq C n^{-1} h^{-d-2k}, \tag{19}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^4 \leq C n^{-2} h^{-2d-4k}. \tag{20}$$

Moreover, for $t \geq 0$,

$$\mathbb{P}\left(\|Z_h\|_{k,\infty;\Omega_h} > C n^{-1/2} h^{-d/2-k} \sqrt{L_h + t} \right) \leq C e^{-t}. \tag{21}$$

Proof. The white-noise isometry gives

> **Reviewer 1**
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> Give intermediate steps.

$$\mathbb{E}|\partial^\beta Z_h(x)|^2 \leq C n^{-1} h^{-d-2|\beta|}.$$

Integrating proves (19). For centered jointly Gaussian variables X, Y, $\mathbb{E}(X^2 Y^2) \leq 3(\mathbb{E}X^2)(\mathbb{E}Y^2)$; double integration proves (20).

The field has a smooth version.

> **Reviewer 1**
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> The following argument is sloppy: give a reference and then argue why a differentiable version with these properties exist; classically this is a Kolmogorov-Centsov Theorem, but there are also specific Gaussian results.

Indeed, after multiplication by a fixed smooth cutoff, its weak derivatives of every order have finite expected squared L^2-norm. Almost surely it therefore belongs to every integer Sobolev space locally. Fourier inversion and Cauchy–Schwarz imply $H^s(\mathbb{R}^d) \hookrightarrow C^k$ for $s > k + d/2$, which gives the assertion.

For a derivative of order k, let $A = n^{-1/2}h^{-d/2-k}$. Its Gaussian increment metric is bounded by

$$CA \min\{1, |x - y|/h\}.$$

On a fixed cube containing Ω_h, use successive grids of mesh $h2^{-j}$. The initial grid has at most Ch^{-d} points, and the j-th grid has at most $Ch^{-d}2^{jd}$ points. Join each point to a nearest point of the preceding grid. Gaussian tail bounds and a union bound show that, outside an event of probability Ce^{-t}, the initial values are bounded by $CA\sqrt{L_h + t}$, and every increment at level $j \geq 1$ is bounded by

$$CA2^{-j}\sqrt{L_h + t + j}.$$

The constants can be chosen so that the failure probabilities at level j sum to Ce^{-t}. Summing the increments and using continuity proves (21). A finite union handles all derivatives of order at most k. Integrating the tail proves (18). □

The next lemma is responsible for the denominator $2\alpha + d + 3$.

Lemma 6.2 (Variance after transport inversion)**.** *Suppose v satisfies* (12) *on Ω_h, and*

$$2d\lambda > \Lambda.$$

If $\|a\|_{1,\infty;\Omega_h} \leq A_0$, then

$$\mathbb{E}\,\|P_v \operatorname{div}(a\nabla Z_h)\|^2_{L^2(\Omega_h)} \leq Cn^{-1}h^{-d-3}. \tag{22}$$

The constant depends only on $d, \lambda, \Lambda, A_0, K$, and not on higher derivatives of v or a.

Proof. For $z \in \Omega_h$, set

$$H_h(z, y) = a(z)\Delta K_h(z - y) + \nabla a(z) \cdot \nabla K_h(z - y).$$

Its $L^2(D)$-norm is at most $Ch^{-d/2-2}$, and its support in y is contained in $B(z, h)$. Consequently,

$$|\langle H_h(z, \cdot), H_h(w, \cdot)\rangle| \leq Ch^{-d-4}\mathbf{1}_{\{|z-w|\leq 2h\}}. \tag{23}$$

Let p be the critical point of v, $c_0 = d\lambda$, and $r = |x - p|$. The representation (14), the white-noise isometry, and (23) give the pointwise variance bound

$$Cn^{-1}h^{-d-4}\int_0^\infty \int_0^\infty e^{-c_0(t+s)}\mathbf{1}_{\{|z_t(x)-z_s(x)|\leq 2h\}}\, ds\, dt.$$

For $s = t + w$, put $r_t = |z_t(x) - p|$. By (13),

> **Reviewer 1**
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> Add details or reference.

$$|z_t - z_{t+w}| \geq r_t - r_{t+w} \geq (1 - e^{-\lambda w}) r_t.$$

If $r_t > 4h$, overlap therefore forces $w \leq 4h/(\lambda r_t)$. If $r_t \leq 4h$, the exponential integral in w is bounded by a constant. Thus

$$\int_0^\infty e^{-c_0 w} \mathbf{1}_{\{|z_t - z_{t+w}| \leq 2h\}} \, dw \leq C \min\{1, h/r_t\}.$$

Since $r_t \geq e^{-\Lambda t} r$, the double time integral is at most

$$C \int_0^\infty e^{-2c_0 t} \min\{1, (h/r) e^{\Lambda t}\} \, dt \leq C \min\{1, h/r\}.$$

The last inequality uses $2c_0 > \Lambda$; at $r = 0$ the minimum is interpreted as 1.

Finally, since $d \geq 2$ and $\Omega_h \subset B(p, 2)$,

$$\int_{\Omega_h} \min\{1, h/|x - p|\} \, dx \leq C \left(h^d + h \int_h^2 r^{d-2} \, dr \right) \leq Ch.$$

Integrating the pointwise variance proves (22). All interchanges of integrals are justified by the exponential weights and the uniform L^2-bound on H_h. □

7 The upper bound

Proposition 7.1. *For sufficiently small $r_* > 0$, the upper bound in Theorem 3.1 holds uniformly over $\mathcal{F}(\alpha, r_*)$.*

Proof. Write

$$p_\alpha = 2\alpha + d + 3, \qquad h = n^{-1/p_\alpha}.$$

Only sufficiently large n will be considered.

Construction of the estimator

Form the observable smooth field

$$\widehat{u}_h(x) = Y_n(K_h(x - \cdot)), \qquad x \in \Omega_h.$$

For a fixed true f, use the deterministic notation

$$u = u_f, \qquad v = S_h u, \qquad a = S_h f.$$

Then $\widehat{u}_h = v + Z_h$.

Choose a fixed $\delta > 0$, depending only on d, so small that

$$\|w - u_\circ\|_{2,\infty;\Omega_h} \leq \delta$$

implies (12), uniformly for $h < 1/8$, with

$$\lambda = \frac{3}{4d}, \qquad \Lambda = \frac{5}{4d}.$$

Indeed, sufficiently small entrywise perturbations of $D^2 u_\circ = I/d$ give these Hessian bounds, and $\nabla u_\circ \cdot \nu = (1 - 2h)/d$ on $\partial\Omega_h$. These constants also satisfy $2d\lambda > \Lambda$.

Polynomial reproduction and Lemma 4.1 imply

$$\|v - u_\circ\|_{2,\infty} \le C r_*.$$

Also, by moving one derivative onto the kernel,

$$\|a\|_{1,\infty} \le C, \qquad \|a\|_{2,\infty} \le C h^{-1}, \qquad \|v\|_{3,\infty} \le C h^{-1}.$$

Decrease r_* so that the first bound is at most $\delta/2$. Choose a fixed sufficiently large M_0, and define the observable event

$$\mathcal{A}_h = \{\|\widehat{u}_h - u_\circ\|_{2,\infty} \le \delta, \quad h\|\widehat{u}_h\|_{3,\infty} \le M_0\}.$$

Lemma 6.1 implies, uniformly in the parameter class,

$$\mathbb{P}_f(\mathcal{A}_h^c) \le C n^{-3}. \tag{24}$$

To verify this explicitly, take $t = 3\log n$ in (21). Both the C^2-noise norm and h times the C^3-noise norm are bounded, with this probability, by a constant times

$$n^{-1/2} h^{-d/2-2} \sqrt{\log n} = h^{\alpha - 1/2} \sqrt{\log n} = o(1).$$

On $\mathcal{A}_h$, define

$$\widetilde{f}_h = P_{\widehat{u}_h} \mathbf{1} \quad \text{on } \Omega_h.$$

Let Π denote pointwise projection onto $[1/2, 3/2]$, and set

$$\widehat{f}_n = \begin{cases} \Pi(E_h \widetilde{f}_h), & \mathcal{A}_h, \\ \mathbf{1}, & \mathcal{A}_h^c. \end{cases}$$

This is an estimator based only on the observations. Its measurability follows from the flow formula: finite-time solutions of the smooth ODE depend continuously on the vector field, and the tail of (14) is uniformly exponentially small on $\mathcal{A}_h$. Notice that the auxiliary transport reconstruction does not require a boundary value for $\widetilde{f}_h$.

Deterministic smoothing bias

Set

$$\mathcal{R}_h = 1 - T_v a.$$

Because $\operatorname{div}(f \nabla u) = 1$, convolution commutes with differentiation in the interior and gives

$$\mathcal{R}_h = -\operatorname{div}\left((S_h f)(S_h \nabla u) - S_h(f \nabla u)\right).$$

Both f and ∇u have uniformly bounded C^α-norms. Lemma 4.2 therefore gives, with

$$B_h = h^\alpha L_h^2,$$

the bounds

$$\|\mathcal{R}_h\|_2 \le C B_h, \qquad \|\mathcal{R}_h\|_{H^1} \le C h^{-1} B_h. \tag{25}$$

Also, Lemmas 4.2 and 4.3 give

$$\|E_h a - f\|_{L^2(D)} \le C h^\alpha L_h. \tag{26}$$

Control of the nonlinear noise contribution

All identities in this paragraph are on $\mathcal{A}_h$. Put

$$e = \widetilde{f}_h - a, \qquad S_Z b = \operatorname{div}(b \nabla Z_h).$$

Since $T_{\widehat{u}_h} = T_v + S_Z$,

$$T_{\widehat{u}_h} e = \mathcal{R}_h - S_Z a. \tag{27}$$

The elementary product bounds

$$\|S_Z a\|_2 \le C\|Z_h\|_{H^2}, \qquad \|S_Z a\|_{H^1} \le C\big(\|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big)$$

follow by differentiating $a\Delta Z_h + \nabla a \cdot \nabla Z_h$ and using the displayed bounds on a. Lemma 5.1, the definition of $\mathcal{A}_h$, and (25) yield the rough estimates

$$\|e\|_2 \le C\big(B_h + \|Z_h\|_{H^2}\big), \tag{28}$$

$$\|e\|_{H^1} \le C\big(h^{-1}B_h + \|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big). \tag{29}$$

To obtain the sharp variance, rewrite (27) as

$$e = P_v\mathcal{R}_h - P_v S_Z a - P_v S_Z e. \tag{30}$$

The middle term is linear in the noise and, by Lemma 6.2,

$$\mathbb{E}\|P_v S_Z a\|_2^2 \le Cn^{-1}h^{-d-3}. \tag{31}$$

The last term is controlled without any independence assumption between the estimated flow and the noise. Namely,

$$\|P_v S_Z e\|_2 \le C\left(\|Z_h\|_{1,\infty}\|e\|_{H^1} + \|Z_h\|_{2,\infty}\|e\|_2\right).$$

Substitute (28)–(29) and apply Lemma 6.1 and Cauchy–Schwarz. This gives

$$\mathbb{E}\left[\mathbf{1}_{\mathcal{A}_h}\|P_v S_Z e\|_2^2\right] \le C\left(B_h^2 n^{-1}h^{-d-4}L_h + n^{-2}h^{-2d-8}L_h\right). \tag{32}$$

For clarity, a representative fourth-moment calculation is

$$\mathbb{E}\left(\|Z_h\|_{1,\infty}^2\|Z_h\|_{H^3}^2\right) \le \left(\mathbb{E}\|Z_h\|_{1,\infty}^4\right)^{1/2}\left(\mathbb{E}\|Z_h\|_{H^3}^4\right)^{1/2} \le Cn^{-2}h^{-2d-8}L_h.$$

The term with $h^{-1}\|Z_h\|_{1,\infty}\|Z_h\|_{H^2}$ has the same bound, as does the term with $\|Z_h\|_{2,\infty}\|Z_h\|_{H^2}$. The terms containing B_h give the first term in (32).

Combining (30)–(32),

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}_h}\|e\|_2^2] \le C\left(B_h^2 + n^{-1}h^{-d-3} + B_h^2 n^{-1}h^{-d-4}L_h + n^{-2}h^{-2d-8}L_h\right).$$

Our bandwidth satisfies $n^{-1} = h^{2\alpha+d+3}$. Hence the last display becomes

$$C\left(h^{2\alpha}L_h^4 + h^{2\alpha} + h^{2\alpha}L_h^4 h^{2\alpha-1}L_h + h^{4\alpha-2}L_h\right) \le Ch^{2\alpha}L_h^4.$$

Here $\alpha > 1$ is used in $h^{4\alpha-2}L_h \le Ch^{2\alpha}$.

Projection onto $[1/2, 3/2]$ cannot increase the loss, since the true conductivity takes values in that interval. By (10), (26), and (24),

$$\sup_{f\in\mathcal{F}(\alpha,r_*)} \mathbb{E}_f\|\widehat{f}_n - f\|_2^2 \le Ch^{2\alpha}L_h^4 + Cn^{-3}.$$

This is the asserted upper bound.

If $\alpha \notin \mathbb{N}$, use the versions of the approximation lemmas without logarithms, so that $B_h = h^\alpha$. The logarithms remaining in the stochastic remainder are absorbed by the strictly positive powers $h^{2\alpha-1}$ and $h^{2\alpha-2}$. The resulting upper bound is $Ch^{2\alpha}$. □

8 The lower bound

We construct pressure perturbations with fine angular oscillations and fixed radial scale.

8.1 Angular bumps

Let $m = d-1$. Fix a smooth coordinate chart $\chi : Q \to \mathbb{S}^{d-1}$, and a smaller cube compactly contained in Q. For sufficiently small h, choose centers $z_1, \dots, z_{M_h}$ in this cube, separated by $4h$, such that

$$ch^{-m} \le M_h \le Ch^{-m}.$$

Fix a nonzero $\psi \in C_c^\infty((-1,1)^m)$, and define

$$\psi_{j,h}(\chi(z)) = \psi((z - z_j)/h),$$

extended by zero outside the chart. Their supports are pairwise disjoint. Coordinate changes on the fixed chart give

$$\|\psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^m, \tag{33}$$

$$\|\nabla_{\mathbb{S}^{d-1}} \psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^{m-2}. \tag{34}$$

Integration by parts and Cauchy–Schwarz imply

$$\|\Delta_{\mathbb{S}^{d-1}} \psi_{j,h}\|_2^2 \ge \frac{\|\nabla_{\mathbb{S}^{d-1}} \psi_{j,h}\|_2^4}{\|\psi_{j,h}\|_2^2} \ge ch^{m-4}. \tag{35}$$

For $\theta \in \{0,1\}^{M_h}$, write $q_\theta = \sum_j \theta_j \psi_{j,h}$. Disjointness gives

$$\|q_\theta\|_2^2 \le C, \qquad \|\nabla^k_{\mathbb{S}^{d-1}} q_\theta\|_\infty \le C_k h^{-k}$$

for every fixed k.

8.2 Realization as conductivities

Fix a nonnegative, nonzero $\rho \in C_c^\infty((1/4, 1/2))$. For a fixed small $\kappa > 0$, define on $\mathbb{R}^d$

$$v_\theta(x) = \kappa h^{\alpha+2} \rho(|x|) q_\theta\left(\frac{x}{|x|}\right), \qquad u_\theta = u_\circ + v_\theta.$$

The perturbation is zero near 0, so this defines a smooth function everywhere. For each fixed k,

$$\|\nabla^k v_\theta\|_\infty \le C_k \kappa h^{\alpha+2-k}. \tag{36}$$

In particular, for sufficiently small κ, $D^2 u_\theta$ is uniformly positive definite, independently of h, θ.

Lemma 8.1 (Conductivity realization and expansion)**.** *For sufficiently small κ, there are smooth positive conductivities $f_\theta = 1 + g_\theta$ such that*

$$\operatorname{div}(f_\theta \nabla u_\theta) = 1 \quad \text{on } \mathbb{R}^d,$$

and, uniformly in h, θ,

$$\|\nabla^k g_\theta\|_\infty \le C_k \kappa h^{\alpha-k} \quad \text{for every fixed } k, \tag{37}$$

$$\|g_\theta\|_{C^\alpha(D)} \le C\kappa, \tag{38}$$

$$g_\theta = \ell_\theta + \varepsilon_\theta, \qquad \ell_\theta = -P_{u_\circ} \Delta v_\theta, \qquad \|\varepsilon_\theta\|_{L^2(D)} \le C\kappa^2 h^{2\alpha}. \tag{39}$$

Moreover, $u_{f_\theta} = u_\theta$ on D.

Proof. The backward-gradient flow of u_θ is globally defined on $\mathbb{R}^d$: its vector field is globally Lipschitz and has at most linear growth. Its unique critical point is 0, and the flow contracts towards 0. Define $f_\theta = P_{u_\theta}\mathbf{1}$ by the same integral as (14). That formula and its proof apply on $\mathbb{R}^d$ here, since the Hessian has uniform positive bounds and all derivatives of order at least two are bounded. Thus f_θ is smooth, positive, bounded, and satisfies the stated equation.

Write $u = u_\theta$, $v = v_\theta$, and $g = f_\theta - 1$. Then

$$T_u g = -\Delta v. \tag{40}$$

The L^∞ transport estimate gives $\|g\|_\infty \le C\kappa h^\alpha$.

We prove (37) by induction. Differentiating (40) k times, and collecting the terms containing k-th derivatives of g, gives

$$T_u(\nabla^k g) + \mathcal{H}_k \nabla^k g = F_k.$$

Here $\mathcal{H}_k$ is the sum of the actions of D^2u on the k tensor indices. If $D^2u \ge \lambda I$, it is symmetric and bounded below by $k\lambda I$. Leibniz's rule gives

$$\begin{aligned}\|F_k\|_\infty \le C_k\Bigg(&\|\nabla^{k+2} v\|_\infty \\ &+ \sum_{j=2}^{k} \|\nabla^{j+1} v\|_\infty \|\nabla^{k-j+1} g\|_\infty \\ &+ \sum_{j=1}^{k} \|\nabla^{j+2} v\|_\infty \|\nabla^{k-j} g\|_\infty \Bigg).\end{aligned}$$

All derivatives of $u_\circ$ of order at least three vanish. Along each trajectory, the homogeneous matrix equation has propagator norm at most $e^{-(d+k)\lambda t}$. Consequently

$$\|\nabla^k g\|_\infty \le ((d+k)\lambda)^{-1} \|F_k\|_\infty.$$

The differentiability and boundedness needed here also follow by the variational-equation argument in Lemma 5.1. Using the induction hypothesis and (36), the right side is bounded by

$$C_k\big(\kappa h^{\alpha-k} + \kappa^2 h^{2\alpha-k}\big) \le C_k' \kappa h^{\alpha-k}.$$

This proves (37).

Choose an integer $J > \alpha$. The dyadic characterization implies

$$\|g\|_{C^\alpha(\mathbb{R}^d)} \le C\left(\|g\|_\infty + \|g\|_\infty^{1-\alpha/J} \|\nabla^J g\|_\infty^{\alpha/J}\right).$$

Indeed, a dyadic block is bounded both by $C\|g\|_\infty$ and, using its vanishing moments, by $C2^{-jJ}\|\nabla^J g\|_\infty$; optimize between these bounds. Substitution of (37) proves (38), including at integer α.

Expanding the original equation around $(1, u_\circ)$ gives

$$T_{u_\circ} g = -\Delta v - \operatorname{div}(g\nabla v).$$

Therefore

$$g = -P_{u_\circ}\Delta v - P_{u_\circ}\operatorname{div}(g\nabla v).$$

The L^∞ bound for $P_{u_\circ}$, together with (36) and (37) at orders zero and one, yields

$$\|P_{u_\circ}\operatorname{div}(g\nabla v)\|_\infty \le C\left(\|\nabla g\|_\infty \|\nabla v\|_\infty + \|g\|_\infty \|\Delta v\|_\infty\right) \le C\kappa^2 h^{2\alpha}.$$

This proves (39).

Finally, v_θ vanishes near ∂D, so $u_\theta = 0$ there. Since $f_\theta > 0$, uniqueness of the Dirichlet problem implies $u_{f_\theta} = u_\theta$. □

8.3 Separation of the conductivities

For $x = r\omega$, the unperturbed inverse is explicitly

$$P_{u_\circ} s(r\omega) = d r^{-d} \int_0^r t^{d-1} s(t\omega)\, dt.$$

Using the polar-coordinate Laplacian, for $r > 1/2$ we obtain

$$\ell_\theta(r\omega) = -d\kappa h^{\alpha+2} J_\rho r^{-d} \Delta_{\mathbb{S}^{d-1}} q_\theta(\omega), \qquad J_\rho = \int_0^\infty t^{d-3} \rho(t)\, dt > 0. \tag{41}$$

Indeed, the radial part integrates to $[t^{d-1}\rho'(t)]_0^r = 0$.

Let $H(\theta, \eta)$ be Hamming distance. Since the angular supports are disjoint, integration over the fixed annulus $3/4 < |x| < 7/8$, using (35) and (41), gives

$$\|\ell_\theta - \ell_\eta\|^2_{L^2(D)} \geq c\kappa^2 h^{2\alpha+m} H(\theta, \eta). \tag{42}$$

There is a subset $\Theta_h \subset \{0,1\}^{M_h}$ such that

$$\log |\Theta_h| \geq c M_h, \qquad H(\theta, \eta) \geq M_h/8 \quad (\theta \neq \eta). \tag{43}$$

This is the usual binary packing bound; see, for example, [4, Lemma 2.9]. One elementary proof is to select words greedily, removing at each step a Hamming ball of radius $\lfloor M_h/8 \rfloor$. Such a ball has at most

$$\sum_{k \leq M_h/8} \binom{M_h}{k} \leq \exp\{M_h H_b(1/8)\}$$

points, where $H_b(t) = -t\log t - (1-t)\log(1-t) < \log 2$. The binomial bound follows by multiplying by $(t/(1-t))^k$ and applying the binomial theorem. The greedy procedure therefore leaves exponentially many selected words and proves (43).

Since $M_h \asymp h^{-m}$, equations (42) and (43) imply

$$\|\ell_\theta - \ell_\eta\|_2 \geq c\kappa h^\alpha \quad (\theta, \eta \in \Theta_h,\ \theta \neq \eta).$$

Lemma 8.1 then gives, for all sufficiently small h,

$$\|f_\theta - f_\eta\|_2 \geq c'\kappa h^\alpha. \tag{44}$$

The remainder is negligible because $\kappa^2 h^{2\alpha} = o(\kappa h^\alpha)$.

8.4 Testing bound

Proposition 8.2. *For every fixed sufficiently small $r > 0$,*

$$R_n(\alpha, r) \geq c\, n^{-2\alpha/(2\alpha+d+3)}$$

for all sufficiently large n.

Proof. Choose a fixed positive κ so small that all the preceding constructions apply and $C\kappa \leq r$, where C is the constant in (38). Thus every f_θ belongs to $\mathcal{F}(\alpha, r)$.

Let P_θ be the observation law under f_θ, and let $P_\circ$ be that under $f = 1$. For Gaussian white noise with variance level n^{-1},

$$\mathrm{KL}(P_\theta, P_\circ) = \frac{n}{2}\|u_\theta - u_\circ\|_2^2 = \frac{n}{2}\|v_\theta\|_2^2 \le Cn\kappa^2 h^{2\alpha+4}. \tag{45}$$

The equality follows directly from the Gaussian likelihood

$$\frac{dP_\theta}{dP_\circ} = \exp\left\{n\big(Y_n(v_\theta) - \langle u_\circ, v_\theta\rangle\big) - \frac{n}{2}\|v_\theta\|_2^2\right\}.$$

It can equivalently be obtained by finite-dimensional projections and convergence of the isonormal series. The inequality in (45) uses $\|q_\theta\|_2^2 \le C$.

Take

$$h = n^{-1/(2\alpha+d+3)}.$$

Since $m = d - 1$,

$$nh^{2\alpha+4} = h^{-m}, \qquad \log|\Theta_h| \ge ch^{-m}.$$

Decrease the fixed κ, if necessary, so that

$$\frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathrm{KL}(P_\theta, P_\circ) \le \frac{1}{8}\log|\Theta_h|. \tag{46}$$

We recall the finite-hypothesis information inequality used here, a form of Fano's inequality; compare [4, Theorem 2.5 and Chapter 2]. Let J be uniform over N hypotheses and let $\overline{P} = N^{-1}\sum_j P_j$. For every reference law P_0,

$$I(J;Y) = \frac{1}{N}\sum_j \mathrm{KL}(P_j, \overline{P}) \le \frac{1}{N}\sum_j \mathrm{KL}(P_j, P_0),$$

because the difference of the right side and the middle expression is $\mathrm{KL}(\overline{P}, P_0)$. For any classifier $\widehat{J}$, with error probability p_e, the binary error indicator gives

$$H(J \mid Y) \le \log 2 + p_e \log N.$$

Since $H(J) = \log N$, it follows that

$$p_e \ge 1 - \frac{I(J;Y) + \log 2}{\log N}.$$

Applying this with (46) shows that every classifier has average error probability at least $1/2$ for large n.

Given any $L^2(D)$-valued estimator, classify by its nearest member of $\{f_\theta : \theta \in \Theta_h\}$, resolving ties by a fixed rule. By (44), misclassification implies an L^2-error of at least $c\kappa h^\alpha/2$. Consequently,

$$\sup_{f\in\mathcal{F}(\alpha,r)} \mathbb{E}_f\|\widehat{f} - f\|_2^2 \ge \frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge c\kappa^2 h^{2\alpha}.$$

Taking the infimum over estimators proves the proposition. □

Proof of Theorem 3.1. Choose r_* sufficiently small for Lemma 4.1 and Proposition 7.1. The upper bound then holds for every subclass $\mathcal{F}(\alpha, r)$, $0 < r \le r_*$. Proposition 8.2, with its fixed κ chosen according to r, gives the matching lower bound. □

Remark 8.3. The effective denominator is $2\alpha + d + 3$, rather than the denominator obtained by treating the forward map as an isotropic operator of a single differential order. In the lower bound there are $d - 1$ angular directions and a two-derivative attenuation. In the upper bound, transport integration reduces the integrated variance of a twice differentiated, smoothed white noise from order $n^{-1}h^{-d-4}$ to order $n^{-1}h^{-d-3}$. The small-radius assumption supplies uniform convexity of the pressure and uniform control of this transport geometry.

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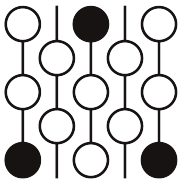
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Review of Problem 2 – Astra

First Proof Third Batch
Reviewer 2
October 8–9, 2026



1 Review

Summary. The manuscript combines the inversion formula for the Darcy problem with a plug-in kernel estimator for the solution field u_f to establish an upper bound on the estimation risk. This is accomplished without a pre-estimation step. On a technical level, the analysis combines properties of kernel estimators with stability properties of the inverse Darcy operator. The lower bound is established using a classical lower-bound strategy based on perturbations of the solution field u_f. The upper and lower bounds match up to logarithmic factors.
Novelty. The general approach to establishing the upper bounds relies on the inversion formula for the Darcy problem combined, as in the human solution. The use of this formula is itself novel in the field. The main difference from the human solution lies in the direct treatment of the L^2-risk, rather than the pointwise risk, and in a different error decomposition. There are also minor differences in the construction of the estimator.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0.5
Justification: The manuscript establishes the correct upper and lower bounds (up to logarithmic factors). While the general estimation and proof methodology is sound and correctly laid out, the necessary details of the proofs are missing. A concrete example is provided by the proof of Lemma 4.3, which is more of a sketch than a rigorous mathematical argument.

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0
Justification: While the paper provides valuable mathematical ideas, it lacks sufficient guidance for the reader. For example, the PDE model is missing. Moreover, the underlying reasons for the non-standard convergence rate are only insufficiently explained in Remark 8.3, at the very end of the manuscript. Some proofs would also benefit from further explanation, in addition to the already mentioned need for more mathematical detail.

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0
Justification: The paper lacks a literature review. Of the four references provided, only one is statistical in nature. Some results are reproved without proper attribution (e.g., the inversion formulas and properties of kernel estimators).

2 Submission with inline comments

Abstract

For a sufficiently small, fixed Hölder–Zygmund ball of conductivities around the constant function 1, the minimax squared L^2-risk is, up to logarithmic factors,

$$n^{-2\alpha/(2\alpha+d+3)}.$$

We prove matching polynomial upper and lower bounds for every $d \geq 2$ and $\alpha > 1$. The upper bound uses a regularized reconstruction along gradient trajectories, together with a variance estimate that exploits integration along these trajectories. The lower bound uses perturbations that oscillate in the $d-1$ angular directions. No boundary condition on the conductivity is imposed.

3 Statement and conventions

Let $D = B(0,1) \subset \mathbb{R}^d$. We interpret $C^s(D)$ as the Hölder–Zygmund restriction space $B^s_{\infty,\infty}(D)$. Any fixed usual equivalent norm may be used; the constants below, including the admissible radius, may depend on this choice.

> **Reviewer 2**
>
> The introduction of the PDE model is missing.

White noise means the isonormal experiment

$$Y_n(\varphi) = \langle u_f, \varphi \rangle_{L^2(D)} + n^{-1/2}\mathbb{W}(\varphi), \qquad \varphi \in L^2(D),$$

where $\mathbb{E}[\mathbb{W}(\varphi)\mathbb{W}(\psi)] = \langle \varphi, \psi \rangle$.

> **Reviewer 2**
>
> Inconsistent notation of inner products.

Write

$$\mathcal{F}(\alpha, r) = \{f : \|f - \mathbf{1}\|_{C^\alpha(D)} \leq r\}, \qquad R_n(\alpha, r) = \inf_{\widehat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\widehat{f} - f\|^2_{L^2(D)}.$$

Theorem 3.1. *Fix $d \geq 2$ and $\alpha > 1$. There exists $r_* > 0$ such that, for each fixed $r \in (0, r_*]$, there are constants $c, C > 0$ and n_0 for which*

$$c\, n^{-2\alpha/(2\alpha+d+3)} \leq R_n(\alpha, r) \leq C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)}, \qquad n \geq n_0.$$

If $\alpha \notin \mathbb{N}$, the logarithmic factor in this upper bound can be omitted. In particular, the answer up to logarithmic factors is

$$\boxed{R_n(\alpha, r) = \mathrel{\widetilde{\asymp}} n^{-2\alpha/(2\alpha+d+3)}.}$$

> **Reviewer 2**
>
> Notation $= \mathrel{\widetilde{\asymp}}$ faulty. Notations for ∂^β and Δ_j (below) not properly defined. The role of a changes throughout the manuscript, often without defining its domain.

For a smooth function a, the notation

$$\|a\|_{k,\infty;\Omega} = \max_{|\beta| \leq k} \|\partial^\beta a\|_{L^\infty(\Omega)}$$

denotes the classical derivative norm, not an integer-index Zygmund norm. Domain subscripts will be suppressed when unambiguous. Constants denoted by C may change from line to line, but are independent of n, the bandwidth, and the conductivity in the specified parameter class.

> **Reviewer 2**
>
> Literature missing.

4 Analytic preliminaries

4.1 Function spaces and the forward equation

We use the Littlewood–Paley characterization

$$\|a\|_{C^s(\mathbb{R}^d)} \asymp \sup_{j\ge 0} 2^{js}\|\Delta_j a\|_\infty \tag{1}$$

for an inhomogeneous smooth dyadic decomposition. The equivalence with classical Hölder spaces at noninteger positive indices, and the differentiation properties of these spaces, are given in [3, Chapter 13, Section 8].

We also use a common bounded extension operator from a smooth bounded domain to $\mathbb{R}^d$, bounded on all the finitely many Besov spaces under consideration. Such an operator exists even for Lipschitz domains; see [2].

Here are some consequences, with the relevant arguments recalled. Differentiation maps C^s boundedly to C^{s-1}. For $s > 0$, C^s is a multiplication algebra:

$$\|ab\|_{C^s} \le C_s\|a\|_{C^s}\|b\|_{C^s}. \tag{2}$$

Indeed, decompose the product into low–high, high–low, and comparable-frequency products. The low–high contribution to its j-th dyadic block is bounded by $C\|a\|_\infty 2^{-js}\|b\|_{C^s}$; interchange a, b for the high–low contribution. For comparable frequencies $k \ge j - C$, the bound is

$$C\|a\|_{C^s}\|b\|_{C^s}\sum_{k\ge j-C} 2^{-2ks} \le C_s 2^{-js}\|a\|_{C^s}\|b\|_{C^s}.$$

This proves (2). Extension and restriction give the same facts on D.

We shall also use the real interpolation identity

$$(C^{s_0}, C^{s_1})_{\theta,\infty} = C^{(1-\theta)s_0+\theta s_1}, \qquad s_0 < s_1, \quad 0 < \theta < 1. \tag{3}$$

For completeness, on $\mathbb{R}^d$ this follows by splitting the dyadic series at a level J. If $s = (1-\theta)s_0 + \theta s_1$, the low and high parts satisfy

$$\|a_{\le J}\|_{C^{s_1}} \le C2^{J(s_1-s)}\|a\|_{C^s}, \qquad \|a_{>J}\|_{C^{s_0}} \le C2^{-J(s-s_0)}\|a\|_{C^s}.$$

Taking $2^{J(s_1-s_0)} \asymp t^{-1}$ bounds the K-functional by $Ct^\theta\|a\|_{C^s}$. Conversely, applying Δ_j to any decomposition $a = a_0 + a_1$ gives

$$\|\Delta_j a\|_\infty \le C2^{-js_0}K(2^{-j(s_1-s_0)}, a; C^{s_0}, C^{s_1}),$$

which proves the reverse bound. A common extension operator transfers the identity to D.

Lemma 4.1 (Uniform perturbative regularity)**.** *Put*

$$u_\circ(x) = \frac{|x|^2 - 1}{2d}.$$

For sufficiently small $r_* > 0$, every $f \in \mathcal{F}(\alpha, r_*)$ satisfies $1/2 \le f \le 3/2$, and the Dirichlet problem

$$\operatorname{div}(f\nabla u) = 1, \qquad u|_{\partial D} = 0$$

has a unique weak solution. Moreover,

$$\|u_f - u_\circ\|_{C^{\alpha+1}(D)} \le C\|f - \mathbf{1}\|_{C^\alpha(D)}. \tag{4}$$

Proof. Let $\mathcal{P}F$ be the zero-Dirichlet solution of $\Delta w = \operatorname{div} F$. For every $s > 1$,

$$\|\mathcal{P}F\|_{C^{s+1}(D)} \le C_s\|F\|_{C^s(D)}. \tag{5}$$

At noninteger s, this is the classical global Schauder estimate and solvability theorem, including higher-order regularity; see [1, Theorems 6.6, 6.14 and 6.19]. At integer s, apply these bounds at two surrounding noninteger indices and use (3). Thus (5) holds also at the Zygmund indices.

Writing $u = u_\circ + w$, the desired equation becomes

$$w = -\mathcal{P}\big((f-1)\nabla u_\circ\big) - \mathcal{P}\big((f-1)\nabla w\big).$$

By (2) and (5), the second term is a linear operator on the zero-trace subspace of $C^{\alpha+1}(D)$ with norm at most $C\|f-1\|_{C^\alpha}$. Choosing r_* so that this norm is at most $1/2$, the Neumann series gives a solution and (4).

The embedding $C^\alpha(D) \hookrightarrow C^0(\overline{D})$ allows us to decrease r_* so that $1/2 \le f \le 3/2$. Uniqueness of a weak solution follows by testing the equation for the difference of two solutions against that difference: $\int_D f|\nabla w|^2 = 0$. Finally, the constructed solution has zero trace and belongs to $H^1_0(D)$, so it is that unique weak solution. □

4.2 Smoothing and a product cancellation

Set

$$m_0 = \lceil \alpha \rceil - 1.$$

Fix a real $K \in C^\infty_c(B(0,1))$ such that

$$\int K = 1, \qquad \int x^\beta K(x)\,dx = 0 \quad (1 \le |\beta| \le 2m_0 + 2). \tag{6}$$

Such a kernel can be obtained from a finite linear combination of distinct dilations of a smooth unit-integral bump: the required coefficients solve a Vandermonde system in the dilation parameters. Positivity of K is neither required nor asserted.

For $0 < h < 1/8$, put

$$K_h(x) = h^{-d}K(x/h), \qquad \Omega_h = B(0, 1-2h), \qquad S_h a(x) = \int_D K_h(x-y)a(y)\,dy, \quad x \in \Omega_h,$$

and let $L_h = \log(e/h)$.

Lemma 4.2 (Smoothing and commutator bounds)**.** *Suppose $a, b \in C^\alpha(D)$ have bounded norms. Uniformly on Ω_h,*

$$\|S_h a - a\|_\infty \le Ch^\alpha L_h, \tag{7}$$

$$\big\|\partial^\gamma\big((S_h a)(S_h b) - S_h(ab)\big)\big\|_\infty \le Ch^{\alpha+1-|\gamma|}L_h^2, \qquad |\gamma| \le 2. \tag{8}$$

If $\alpha \notin \mathbb{N}$, all factors L_h in these two bounds may be replaced by 1. The statements hold componentwise for vector-valued b.

Proof. We first record the local Taylor remainder

$$\sup_{|y-x|\le h} |a(y) - P_{a,x}(y)| \le Ch^\alpha L_h, \tag{9}$$

where $P_{a,x}$ is the Taylor polynomial of degree m_0. Its coefficients are uniformly bounded. At noninteger α, this is the ordinary Hölder Taylor estimate, without L_h. At integer α, derivatives of order $\alpha - 1$ have modulus of continuity bounded by $Ct\log(e/t)$. To see the latter directly from (1), split their dyadic series at $2^J \asymp t^{-1}$: the low-frequency increments sum to $C\sum_{j\le J} t$, and the high-frequency tail is at most $C\sum_{j>J} 2^{-j}$. Taylor's integral remainder then gives (9). Extension to $\mathbb{R}^d$ justifies this argument uniformly near the boundary.

Polynomial reproduction from (6) immediately proves (7). To prove (8), write

$$Q_h(a,b) = (S_h a)(S_h b) - S_h(ab).$$

Fix x and write $a = P_a + R_a$, $b = P_b + R_b$, using the Taylor polynomials at x. Keep these polynomials fixed when differentiating at x. Polynomial reproduction gives $Q_h(P_a, P_b) = 0$. Also $Q_h(a, c) = 0$ for every constant c.

On $B(x,h)$, the remainders satisfy $|R_a|+|R_b| \le Ch^\alpha L_h$, whereas $|P_a - P_a(x)|+|P_b - P_b(x)| \le Ch$. Differentiating the convolution kernels gives, for $|\gamma| \le 2$,

$$|\partial^\gamma Q_h(R_a, P_b)(x)| + |\partial^\gamma Q_h(P_a, R_b)(x)| \le Ch^{\alpha+1-|\gamma|} L_h.$$

For example, in the first expression subtract the constant $P_b(x)$; the differentiated convolution of $R_a(P_b - P_b(x))$ has precisely this bound. In the product term, a derivative falling on the polynomial is bounded by Ch^{1-k} at derivative order $k \ge 1$; the undifferentiated polynomial vanishes at x. Finally,

$$|\partial^\gamma Q_h(R_a, R_b)(x)| \le Ch^{2\alpha-|\gamma|} L_h^2.$$

Since $\alpha > 1$, these inequalities prove (8). The same proof without logarithms applies at noninteger α. □

Lemma 4.3 (Completion of a thin boundary strip)**.** *For all sufficiently small h, there is a linear map $E_h : L^2(\Omega_h) \to L^2(D)$ such that*

$$\|E_h a\|_{L^2(D)} \le C\|a\|_{L^2(\Omega_h)}, \tag{10}$$

$$\|E_h(f|_{\Omega_h}) - f\|_{L^2(D)} \le C\|f\|_{C^\alpha(D)} h^\alpha L_h. \tag{11}$$

At noninteger α, L_h can be omitted.

Proof. Let $q = m_0 + 1$ and $\lambda_j = (-1)^{j+1}\binom{q}{j}$, $1 \le j \le q$. Then

$$\sum_{j=1}^q \lambda_j = 1, \qquad \sum_{j=1}^q \lambda_j j^k = 0, \quad 1 \le k \le m_0.$$

Define $E_h a = a$ on Ω_h, and, for $x \in D \setminus \Omega_h$, define

$$E_h a(x) = \sum_{j=1}^q \lambda_j a((1-3jh)x).$$

For $h < 1/(6q)$, all arguments on the right belong to Ω_h and all dilation factors are at least $1/2$. A change of variables proves (10). Taylor expansion of f at x, together with the displayed moment identities and (9), proves (11). □

5 Inversion along gradient trajectories

For a smooth function v, define

$$T_v a = \operatorname{div}(a\nabla v) = \nabla v \cdot \nabla a + (\Delta v)a.$$

Lemma 5.1 (Transport inverse)**.** *Let $\Omega = B(0,R)$, $3/4 \le R \le 1$, and suppose v is smooth on a neighborhood of $\overline{\Omega}$, with*

$$\lambda I \le D^2 v \le \Lambda I, \qquad \nabla v \cdot \nu > 0 \text{ on } \partial\Omega, \tag{12}$$

where $0 < \lambda \le \Lambda$. There is a unique critical point $p \in \Omega$. Let $z_t(x)$ solve

$$\frac{d}{dt} z_t(x) = -\nabla v(z_t(x)), \qquad z_0(x) = x.$$

This flow exists in $\overline{\Omega}$ for all $t \ge 0$, and

$$e^{-\Lambda t}|x-p| \le |z_t(x) - p| \le e^{-\lambda t}|x-p|. \tag{13}$$

For smooth g, the unique bounded smooth solution of $T_v a = g$ is

$$P_v g(x) = \int_0^\infty \exp\left\{-\int_0^t \Delta v(z_s(x))\, ds\right\} g(z_t(x))\, dt. \tag{14}$$

With $c_0 = d\lambda$, it satisfies

$$\|P_v g\|_\infty \le c_0^{-1}\|g\|_\infty, \tag{15}$$

$$\|P_v g\|_2 \le 2c_0^{-1}\|g\|_2, \tag{16}$$

$$\|P_v g\|_{H^1} \le C\big\{\|g\|_{H^1} + (1 + \|D^3 v\|_\infty)\|g\|_2\big\}. \tag{17}$$

The constants in the last estimate depend only on d, λ, Λ.

> **Reviewer 2**
>
> Attribution for the inversion formula missing.

Proof. Strict convexity gives at most one critical point. A minimum on $\overline{\Omega}$ cannot occur at the boundary, because the inward direction decreases v, so this critical point exists in Ω. The vector field $-\nabla v$ points inward at the boundary; the usual first-exit argument

> **Reviewer 2**
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> Details or reference required.

and local existence for a smooth vector field therefore give a global backward-gradient flow in $\overline{\Omega}$. For $z \in \Omega$,

$$\lambda|z-p|^2 \le (z-p)\cdot\nabla v(z) \le \Lambda|z-p|^2.$$

Applying this to the derivative of $|z_t - p|$ proves (13).

Since $\Delta v \ge c_0$, the integral in (14) converges absolutely and gives (15). It is smooth up to the boundary. Here is a justification of differentiating the infinite-time integral. The first variational equation gives $\|D_x z_t\| \le e^{-\lambda t}$. Inductively, the equation for $D_x^k z_t$ has homogeneous part $-D^2 v(z_t) D_x^k z_t$, while each forcing term is a product of at least two lower-order flow derivatives. Variation of constants therefore gives $\|D_x^k z_t\| \le C_k e^{-\lambda t}$. Derivatives of $\int_0^t \Delta v(z_s)\, ds$ of every positive order are consequently bounded uniformly in t. Every differentiated integrand is thus bounded by $C_k e^{-c_0 t}$.

> **Reviewer 2**
>
> Argument is lacking mathematical precision and more details should be given. This issue persists throughout the manuscript.

The flow semigroup property, or differentiation after splitting the integral at a finite time, gives $T_v P_v g = g$. If $T_v a = 0$, differentiation along the flow shows

$$a(x) = \exp\left\{-\int_0^t \Delta v(z_s(x))\, ds\right\} a(z_t(x)).$$

Letting $t \to \infty$ proves uniqueness among bounded solutions.

For $a = P_v g$, integration by parts gives

$$\int_\Omega a T_v a = \frac{1}{2}\int_{\partial\Omega} (\nabla v \cdot \nu) a^2 + \frac{1}{2}\int_\Omega (\Delta v) a^2 \geq \frac{c_0}{2}\|a\|_2^2.$$

This proves (16). Differentiating the equation gives

$$T_v(\nabla a) + D^2 v\, \nabla a = \nabla g - a \nabla \Delta v.$$

The same integration by parts, now for a vector field, yields

$$(c_0/2 + \lambda)\|\nabla a\|_2 \leq \|\nabla g\|_2 + \|\nabla \Delta v\|_\infty \|a\|_2.$$

Together with (16), this proves (17). □

6 Gaussian smoothing and the decisive variance estimate

Define

$$Z_h(x) = n^{-1/2}\mathbb{W}(K_h(x - \cdot)), \qquad x \in \mathbb{R}^d,$$

where the white-noise argument is restricted to D.

Lemma 6.1 (Gaussian smoothing bounds)**.** *For every fixed nonnegative integer k and $p \geq 1$,*

$$\mathbb{E}\|Z_h\|_{k,\infty;\Omega_h}^p \leq C_p n^{-p/2} h^{-p(d/2+k)} L_h^{p/2}, \tag{18}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^2 \leq C n^{-1} h^{-d-2k}, \tag{19}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^4 \leq C n^{-2} h^{-2d-4k}. \tag{20}$$

Moreover, for $t \geq 0$,

$$\mathbb{P}\left(\|Z_h\|_{k,\infty;\Omega_h} > C n^{-1/2} h^{-d/2-k}\sqrt{L_h + t}\right) \leq C e^{-t}. \tag{21}$$

Proof. The white-noise isometry gives

$$\mathbb{E}|\partial^\beta Z_h(x)|^2 \leq C n^{-1} h^{-d-2|\beta|}.$$

Integrating proves (19). For centered jointly Gaussian variables X, Y, $\mathbb{E}(X^2Y^2) \leq 3(\mathbb{E}X^2)(\mathbb{E}Y^2)$; double integration proves (20).

The field has a smooth version. Indeed, after multiplication by a fixed smooth cutoff, its weak derivatives of every order have finite expected squared L^2-norm. Almost surely it therefore belongs

to every integer Sobolev space locally. Fourier inversion and Cauchy–Schwarz imply $H^s(\mathbb{R}^d) \hookrightarrow C^k$ for $s > k + d/2$, which gives the assertion.

For a derivative of order k, let $A = n^{-1/2}h^{-d/2-k}$. Its Gaussian increment metric is bounded by

$$CA \min\{1, |x-y|/h\}.$$

On a fixed cube containing Ω_h, use successive grids of mesh $h2^{-j}$. The initial grid has at most Ch^{-d} points, and the j-th grid has at most $Ch^{-d}2^{jd}$ points. Join each point to a nearest point of the preceding grid. Gaussian tail bounds and a union bound show that, outside an event of probability Ce^{-t}, the initial values are bounded by $CA\sqrt{L_h + t}$, and every increment at level $j \geq 1$ is bounded by

$$CA2^{-j}\sqrt{L_h + t + j}.$$

The constants can be chosen so that the failure probabilities at level j sum to Ce^{-t}. Summing the increments and using continuity proves (21). A finite union handles all derivatives of order at most k. Integrating the tail proves (18). □

The next lemma is responsible for the denominator $2\alpha + d + 3$.

Lemma 6.2 (Variance after transport inversion)**.** *Suppose v satisfies* (12) *on Ω_h, and*

$$2d\lambda > \Lambda.$$

If $\|a\|_{1,\infty;\Omega_h} \leq A_0$, then

$$\mathbb{E}\,\|P_v \operatorname{div}(a\nabla Z_h)\|^2_{L^2(\Omega_h)} \leq Cn^{-1}h^{-d-3}. \tag{22}$$

The constant depends only on $d, \lambda, \Lambda, A_0, K$, and not on higher derivatives of v or a.

Proof. For $z \in \Omega_h$, set

$$H_h(z,y) = a(z)\Delta K_h(z-y) + \nabla a(z) \cdot \nabla K_h(z-y).$$

Its $L^2(D)$-norm is at most $Ch^{-d/2-2}$, and its support in y is contained in $B(z,h)$. Consequently,

$$|\langle H_h(z,\cdot), H_h(w,\cdot)\rangle| \leq Ch^{-d-4}\mathbf{1}_{\{|z-w|\leq 2h\}}. \tag{23}$$

Let p be the critical point of v, $c_0 = d\lambda$, and $r = |x-p|$. The representation (14), the white-noise isometry, and (23) give the pointwise variance bound

$$Cn^{-1}h^{-d-4}\int_0^\infty \int_0^\infty e^{-c_0(t+s)}\mathbf{1}_{\{|z_t(x)-z_s(x)|\leq 2h\}}\,ds\,dt.$$

For $s = t + w$, put $r_t = |z_t(x) - p|$. By (13),

$$|z_t - z_{t+w}| \geq r_t - r_{t+w} \geq (1 - e^{-\lambda w})r_t.$$

If $r_t > 4h$, overlap therefore forces $w \leq 4h/(\lambda r_t)$. If $r_t \leq 4h$, the exponential integral in w is bounded by a constant. Thus

$$\int_0^\infty e^{-c_0 w}\mathbf{1}_{\{|z_t - z_{t+w}|\leq 2h\}}\,dw \leq C\min\{1, h/r_t\}.$$

Since $r_t \geq e^{-\Lambda t}r$, the double time integral is at most

$$C\int_0^\infty e^{-2c_0 t}\min\{1, (h/r)e^{\Lambda t}\}\,dt \leq C\min\{1, h/r\}.$$

The last inequality uses $2c_0 > \Lambda$; at $r = 0$ the minimum is interpreted as 1.

Finally, since $d \geq 2$ and $\Omega_h \subset B(p, 2)$,

$$\int_{\Omega_h} \min\{1, h/|x-p|\}\, dx \leq C\left(h^d + h\int_h^2 r^{d-2}\, dr\right) \leq Ch.$$

Integrating the pointwise variance proves (22). All interchanges of integrals are justified by the exponential weights and the uniform L^2-bound on H_h. □

7 The upper bound

Proposition 7.1. *For sufficiently small $r_* > 0$, the upper bound in Theorem 3.1 holds uniformly over $\mathcal{F}(\alpha, r_*)$.*

Proof. Write

$$p_\alpha = 2\alpha + d + 3, \qquad h = n^{-1/p_\alpha}.$$

Only sufficiently large n will be considered.

Construction of the estimator

Form the observable smooth field

$$\widehat{u}_h(x) = Y_n(K_h(x - \cdot)), \qquad x \in \Omega_h.$$

For a fixed true f, use the deterministic notation

$$u = u_f, \qquad v = S_h u, \qquad a = S_h f.$$

Then $\widehat{u}_h = v + Z_h$.

Choose a fixed $\delta > 0$, depending only on d, so small that

$$\|w - u_\circ\|_{2,\infty;\Omega_h} \leq \delta$$

implies (12), uniformly for $h < 1/8$, with

$$\lambda = \frac{3}{4d}, \qquad \Lambda = \frac{5}{4d}.$$

Indeed, sufficiently small entrywise perturbations of $D^2 u_\circ = I/d$ give these Hessian bounds, and $\nabla u_\circ \cdot \nu = (1-2h)/d$ on $\partial\Omega_h$. These constants also satisfy $2d\lambda > \Lambda$.

Polynomial reproduction and Lemma 4.1 imply

$$\|v - u_\circ\|_{2,\infty} \leq Cr_*.$$

Also, by moving one derivative onto the kernel,

$$\|a\|_{1,\infty} \leq C, \qquad \|a\|_{2,\infty} \leq Ch^{-1}, \qquad \|v\|_{3,\infty} \leq Ch^{-1}.$$

Decrease r_* so that the first bound is at most $\delta/2$. Choose a fixed sufficiently large M_0, and define the observable event

$$\mathcal{A}_h = \left\{\|\widehat{u}_h - u_\circ\|_{2,\infty} \leq \delta, \quad h\|\widehat{u}_h\|_{3,\infty} \leq M_0\right\}.$$

Lemma 6.1 implies, uniformly in the parameter class,

$$\mathbb{P}_f(\mathcal{A}_h^c) \le Cn^{-3}. \tag{24}$$

To verify this explicitly, take $t = 3\log n$ in (21). Both the C^2-noise norm and h times the C^3-noise norm are bounded, with this probability, by a constant times

$$n^{-1/2}h^{-d/2-2}\sqrt{\log n} = h^{\alpha-1/2}\sqrt{\log n} = o(1).$$

On $\mathcal{A}_h$, define

$$\widetilde{f}_h = P_{\widehat{u}_h}\mathbf{1} \quad \text{on } \Omega_h.$$

Let Π denote pointwise projection onto $[1/2, 3/2]$,

> **Reviewer 2**
>
> The reason for the projection should be given.

and set

$$\widehat{f}_n = \begin{cases} \Pi(E_h\widetilde{f}_h), & \mathcal{A}_h, \\ \mathbf{1}, & \mathcal{A}_h^c. \end{cases}$$

This is an estimator based only on the observations. Its measurability follows from the flow formula: finite-time solutions of the smooth ODE depend continuously on the vector field, and the tail of (14) is uniformly exponentially small on $\mathcal{A}_h$. Notice that the auxiliary transport reconstruction does not require a boundary value for $\widetilde{f}_h$.

> **Reviewer 2**
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> The following paragraphs discuss the error decomposition, which should be given here for the reader's convenience.

Deterministic smoothing bias

Set

$$\mathcal{R}_h = 1 - T_v a.$$

Because $\operatorname{div}(f\nabla u) = 1$, convolution commutes with differentiation in the interior and gives

$$\mathcal{R}_h = -\operatorname{div}\left((S_h f)(S_h\nabla u) - S_h(f\nabla u)\right).$$

Both f and ∇u have uniformly bounded C^α-norms. Lemma 4.2 therefore gives, with

$$B_h = h^\alpha L_h^2,$$

the bounds

$$\|\mathcal{R}_h\|_2 \le CB_h, \qquad \|\mathcal{R}_h\|_{H^1} \le Ch^{-1}B_h. \tag{25}$$

Also, Lemmas 4.2 and 4.3 give

$$\|E_h a - f\|_{L^2(D)} \le Ch^\alpha L_h. \tag{26}$$

Control of the nonlinear noise contribution

All identities in this paragraph are on $\mathcal{A}_h$. Put

$$e = \widetilde{f}_h - a, \qquad S_Z b = \operatorname{div}(b \nabla Z_h).$$

Since $T_{\widehat{u}_h} = T_v + S_Z$,

$$T_{\widehat{u}_h} e = \mathcal{R}_h - S_Z a. \tag{27}$$

The elementary product bounds

$$\|S_Z a\|_2 \le C\|Z_h\|_{H^2}, \qquad \|S_Z a\|_{H^1} \le C\big(\|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big)$$

follow by differentiating $a\Delta Z_h + \nabla a \cdot \nabla Z_h$ and using the displayed bounds on a. Lemma 5.1, the definition of $\mathcal{A}_h$, and (25) yield the rough estimates

$$\|e\|_2 \le C\big(B_h + \|Z_h\|_{H^2}\big), \tag{28}$$

$$\|e\|_{H^1} \le C\big(h^{-1}B_h + \|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big). \tag{29}$$

To obtain the sharp variance, rewrite (27) as

$$e = P_v\mathcal{R}_h - P_v S_Z a - P_v S_Z e. \tag{30}$$

The middle term is linear in the noise and, by Lemma 6.2,

$$\mathbb{E}\|P_v S_Z a\|_2^2 \le Cn^{-1}h^{-d-3}. \tag{31}$$

The last term is controlled without any independence assumption between the estimated flow and the noise. Namely,

$$\|P_v S_Z e\|_2 \le C\left(\|Z_h\|_{1,\infty}\|e\|_{H^1} + \|Z_h\|_{2,\infty}\|e\|_2\right).$$

Substitute (28)–(29) and apply Lemma 6.1 and Cauchy–Schwarz. This gives

$$\mathbb{E}\left[\mathbf{1}_{\mathcal{A}_h}\|P_v S_Z e\|_2^2\right] \le C\left(B_h^2 n^{-1}h^{-d-4}L_h + n^{-2}h^{-2d-8}L_h\right). \tag{32}$$

For clarity, a representative fourth-moment calculation is

$$\mathbb{E}\left(\|Z_h\|_{1,\infty}^2\|Z_h\|_{H^3}^2\right) \le \left(\mathbb{E}\|Z_h\|_{1,\infty}^4\right)^{1/2}\left(\mathbb{E}\|Z_h\|_{H^3}^4\right)^{1/2} \le Cn^{-2}h^{-2d-8}L_h.$$

The term with $h^{-1}\|Z_h\|_{1,\infty}\|Z_h\|_{H^2}$ has the same bound, as does the term with $\|Z_h\|_{2,\infty}\|Z_h\|_{H^2}$. The terms containing B_h give the first term in (32).

Combining (30)–(32),

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}_h}\|e\|_2^2] \le C\left(B_h^2 + n^{-1}h^{-d-3} + B_h^2 n^{-1}h^{-d-4}L_h + n^{-2}h^{-2d-8}L_h\right).$$

Our bandwidth satisfies $n^{-1} = h^{2\alpha+d+3}$. Hence the last display becomes

$$C\left(h^{2\alpha}L_h^4 + h^{2\alpha} + h^{2\alpha}L_h^4 h^{2\alpha-1}L_h + h^{4\alpha-2}L_h\right) \le Ch^{2\alpha}L_h^4.$$

Here $\alpha > 1$ is used in $h^{4\alpha-2}L_h \le Ch^{2\alpha}$.

Projection onto $[1/2, 3/2]$ cannot increase the loss, since the true conductivity takes values in that interval. By (10), (26), and (24),

$$\sup_{f\in\mathcal{F}(\alpha,r_*)} \mathbb{E}_f\|\widehat{f}_n - f\|_2^2 \le Ch^{2\alpha}L_h^4 + Cn^{-3}.$$

This is the asserted upper bound.

If $\alpha \notin \mathbb{N}$, use the versions of the approximation lemmas without logarithms, so that $B_h = h^\alpha$. The logarithms remaining in the stochastic remainder are absorbed by the strictly positive powers $h^{2\alpha-1}$ and $h^{2\alpha-2}$. The resulting upper bound is $Ch^{2\alpha}$. □

8 The lower bound

We construct pressure perturbations with fine angular oscillations and fixed radial scale.

8.1 Angular bumps

Let $m = d-1$. Fix a smooth coordinate chart $\chi : Q \to \mathbb{S}^{d-1}$, and a smaller cube compactly contained in Q. For sufficiently small h, choose centers $z_1, \dots, z_{M_h}$ in this cube, separated by $4h$, such that

$$ch^{-m} \le M_h \le Ch^{-m}.$$

Fix a nonzero $\psi \in C_c^\infty((-1,1)^m)$, and define

$$\psi_{j,h}(\chi(z)) = \psi((z - z_j)/h),$$

extended by zero outside the chart. Their supports are pairwise disjoint. Coordinate changes on the fixed chart give

$$\|\psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^m, \tag{33}$$

$$\|\nabla_{\mathbb{S}^{d-1}} \psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^{m-2}. \tag{34}$$

Integration by parts and Cauchy–Schwarz imply

$$\|\Delta_{\mathbb{S}^{d-1}} \psi_{j,h}\|_2^2 \ge \frac{\|\nabla_{\mathbb{S}^{d-1}} \psi_{j,h}\|_2^4}{\|\psi_{j,h}\|_2^2} \ge ch^{m-4}. \tag{35}$$

For $\theta \in \{0,1\}^{M_h}$, write $q_\theta = \sum_j \theta_j \psi_{j,h}$. Disjointness gives

$$\|q_\theta\|_2^2 \le C, \qquad \|\nabla^k_{\mathbb{S}^{d-1}} q_\theta\|_\infty \le C_k h^{-k}$$

for every fixed k.

8.2 Realization as conductivities

Fix a nonnegative, nonzero $\rho \in C_c^\infty((1/4, 1/2))$. For a fixed small $\kappa > 0$, define on $\mathbb{R}^d$

$$v_\theta(x) = \kappa h^{\alpha+2} \rho(|x|) q_\theta\left(\frac{x}{|x|}\right), \qquad u_\theta = u_\circ + v_\theta.$$

The perturbation is zero near 0, so this defines a smooth function everywhere. For each fixed k,

$$\|\nabla^k v_\theta\|_\infty \le C_k \kappa h^{\alpha+2-k}. \tag{36}$$

In particular, for sufficiently small κ, $D^2 u_\theta$ is uniformly positive definite, independently of h, θ.

Lemma 8.1 (Conductivity realization and expansion)**.** *For sufficiently small κ, there are smooth positive conductivities $f_\theta = 1 + g_\theta$ such that*

$$\operatorname{div}(f_\theta \nabla u_\theta) = 1 \quad \text{on } \mathbb{R}^d,$$

and, uniformly in h, θ,

$$\|\nabla^k g_\theta\|_\infty \le C_k \kappa h^{\alpha-k} \quad \text{for every fixed } k, \tag{37}$$

$$\|g_\theta\|_{C^\alpha(D)} \le C\kappa, \tag{38}$$

$$g_\theta = \ell_\theta + \varepsilon_\theta, \qquad \ell_\theta = -P_{u_\circ} \Delta v_\theta, \qquad \|\varepsilon_\theta\|_{L^2(D)} \le C\kappa^2 h^{2\alpha}. \tag{39}$$

Moreover, $u_{f_\theta} = u_\theta$ on D.

Proof. The backward-gradient flow of u_θ is globally defined on $\mathbb{R}^d$: its vector field is globally Lipschitz and has at most linear growth. Its unique critical point is 0, and the flow contracts towards 0. Define $f_\theta = P_{u_\theta}\mathbf{1}$ by the same integral as (14). That formula and its proof apply on $\mathbb{R}^d$ here, since the Hessian has uniform positive bounds and all derivatives of order at least two are bounded. Thus f_θ is smooth, positive, bounded, and satisfies the stated equation.

Write $u = u_\theta$, $v = v_\theta$, and $g = f_\theta - 1$. Then

$$T_u g = -\Delta v. \tag{40}$$

The L^∞ transport estimate gives $\|g\|_\infty \le C\kappa h^\alpha$.

We prove (37) by induction. Differentiating (40) k times, and collecting the terms containing k-th derivatives of g, gives

$$T_u(\nabla^k g) + \mathcal{H}_k \nabla^k g = F_k.$$

Here $\mathcal{H}_k$ is the sum of the actions of D^2u on the k tensor indices. If $D^2u \ge \lambda I$, it is symmetric and bounded below by $k\lambda I$. Leibniz's rule gives

$$\begin{aligned}\|F_k\|_\infty \le C_k\Bigg(&\|\nabla^{k+2} v\|_\infty \\ &+ \sum_{j=2}^{k} \|\nabla^{j+1} v\|_\infty \|\nabla^{k-j+1} g\|_\infty \\ &+ \sum_{j=1}^{k} \|\nabla^{j+2} v\|_\infty \|\nabla^{k-j} g\|_\infty \Bigg).\end{aligned}$$

All derivatives of $u_\circ$ of order at least three vanish. Along each trajectory, the homogeneous matrix equation has propagator norm at most $e^{-(d+k)\lambda t}$. Consequently

$$\|\nabla^k g\|_\infty \le ((d+k)\lambda)^{-1} \|F_k\|_\infty.$$

The differentiability and boundedness needed here also follow by the variational-equation argument in Lemma 5.1. Using the induction hypothesis and (36), the right side is bounded by

$$C_k\big(\kappa h^{\alpha-k} + \kappa^2 h^{2\alpha-k}\big) \le C_k' \kappa h^{\alpha-k}.$$

This proves (37).

Choose an integer $J > \alpha$. The dyadic characterization implies

$$\|g\|_{C^\alpha(\mathbb{R}^d)} \le C\left(\|g\|_\infty + \|g\|_\infty^{1-\alpha/J} \|\nabla^J g\|_\infty^{\alpha/J}\right).$$

Indeed, a dyadic block is bounded both by $C\|g\|_\infty$ and, using its vanishing moments, by $C2^{-jJ}\|\nabla^J g\|_\infty$; optimize between these bounds. Substitution of (37) proves (38), including at integer α.

Expanding the original equation around $(1, u_\circ)$ gives

$$T_{u_\circ} g = -\Delta v - \operatorname{div}(g\nabla v).$$

Therefore

$$g = -P_{u_\circ}\Delta v - P_{u_\circ}\operatorname{div}(g\nabla v).$$

The L^∞ bound for $P_{u_\circ}$, together with (36) and (37) at orders zero and one, yields

$$\|P_{u_\circ}\operatorname{div}(g\nabla v)\|_\infty \le C\left(\|\nabla g\|_\infty \|\nabla v\|_\infty + \|g\|_\infty \|\Delta v\|_\infty\right) \le C\kappa^2 h^{2\alpha}.$$

This proves (39).

Finally, v_θ vanishes near ∂D, so $u_\theta = 0$ there. Since $f_\theta > 0$, uniqueness of the Dirichlet problem implies $u_{f_\theta} = u_\theta$. □

8.3 Separation of the conductivities

For $x = r\omega$, the unperturbed inverse is explicitly

$$P_{u_\circ} s(r\omega) = d r^{-d} \int_0^r t^{d-1} s(t\omega)\, dt.$$

Using the polar-coordinate Laplacian, for $r > 1/2$ we obtain

$$\ell_\theta(r\omega) = -d\kappa h^{\alpha+2} J_\rho r^{-d} \Delta_{\mathbb{S}^{d-1}} q_\theta(\omega), \qquad J_\rho = \int_0^\infty t^{d-3} \rho(t)\, dt > 0. \tag{41}$$

Indeed, the radial part integrates to $[t^{d-1}\rho'(t)]_0^r = 0$.

Let $H(\theta, \eta)$ be Hamming distance. Since the angular supports are disjoint, integration over the fixed annulus $3/4 < |x| < 7/8$, using (35) and (41), gives

$$\|\ell_\theta - \ell_\eta\|^2_{L^2(D)} \geq c\kappa^2 h^{2\alpha+m} H(\theta, \eta). \tag{42}$$

There is a subset $\Theta_h \subset \{0,1\}^{M_h}$ such that

$$\log |\Theta_h| \geq c M_h, \qquad H(\theta, \eta) \geq M_h/8 \quad (\theta \neq \eta). \tag{43}$$

This is the usual binary packing bound; see, for example, [4, Lemma 2.9]. One elementary proof is to select words greedily, removing at each step a Hamming ball of radius $\lfloor M_h/8 \rfloor$. Such a ball has at most

$$\sum_{k \leq M_h/8} \binom{M_h}{k} \leq \exp\{M_h H_b(1/8)\}$$

points, where $H_b(t) = -t\log t - (1-t)\log(1-t) < \log 2$. The binomial bound follows by multiplying by $(t/(1-t))^k$ and applying the binomial theorem. The greedy procedure therefore leaves exponentially many selected words and proves (43).

Since $M_h \asymp h^{-m}$, equations (42) and (43) imply

$$\|\ell_\theta - \ell_\eta\|_2 \geq c\kappa h^\alpha \quad (\theta, \eta \in \Theta_h,\ \theta \neq \eta).$$

Lemma 8.1 then gives, for all sufficiently small h,

$$\|f_\theta - f_\eta\|_2 \geq c'\kappa h^\alpha. \tag{44}$$

The remainder is negligible because $\kappa^2 h^{2\alpha} = o(\kappa h^\alpha)$.

8.4 Testing bound

Proposition 8.2. *For every fixed sufficiently small $r > 0$,*

$$R_n(\alpha, r) \geq c\, n^{-2\alpha/(2\alpha+d+3)}$$

for all sufficiently large n.

Proof. Choose a fixed positive κ so small that all the preceding constructions apply and $C\kappa \leq r$, where C is the constant in (38). Thus every f_θ belongs to $\mathcal{F}(\alpha, r)$.

Let P_θ be the observation law under f_θ, and let $P_\circ$ be that under $f = 1$. For Gaussian white noise with variance level n^{-1},

$$\mathrm{KL}(P_\theta, P_\circ) = \frac{n}{2}\|u_\theta - u_\circ\|_2^2 = \frac{n}{2}\|v_\theta\|_2^2 \le Cn\kappa^2 h^{2\alpha+4}. \tag{45}$$

The equality follows directly from the Gaussian likelihood

$$\frac{dP_\theta}{dP_\circ} = \exp\left\{n\big(Y_n(v_\theta) - \langle u_\circ, v_\theta\rangle\big) - \frac{n}{2}\|v_\theta\|_2^2\right\}.$$

It can equivalently be obtained by finite-dimensional projections and convergence of the isonormal series. The inequality in (45) uses $\|q_\theta\|_2^2 \le C$.

Take

$$h = n^{-1/(2\alpha+d+3)}.$$

Since $m = d - 1$,

$$nh^{2\alpha+4} = h^{-m}, \qquad \log|\Theta_h| \ge ch^{-m}.$$

Decrease the fixed κ, if necessary, so that

$$\frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathrm{KL}(P_\theta, P_\circ) \le \frac{1}{8}\log|\Theta_h|. \tag{46}$$

We recall the finite-hypothesis information inequality used here, a form of Fano's inequality; compare [4, Theorem 2.5 and Chapter 2]. Let J be uniform over N hypotheses and let $\overline{P} = N^{-1}\sum_j P_j$. For every reference law P_0,

$$I(J;Y) = \frac{1}{N}\sum_j \mathrm{KL}(P_j, \overline{P}) \le \frac{1}{N}\sum_j \mathrm{KL}(P_j, P_0),$$

because the difference of the right side and the middle expression is $\mathrm{KL}(\overline{P}, P_0)$. For any classifier $\widehat{J}$, with error probability p_e, the binary error indicator gives

$$H(J \mid Y) \le \log 2 + p_e \log N.$$

Since $H(J) = \log N$, it follows that

$$p_e \ge 1 - \frac{I(J;Y) + \log 2}{\log N}.$$

Applying this with (46) shows that every classifier has average error probability at least $1/2$ for large n.

Given any $L^2(D)$-valued estimator, classify by its nearest member of $\{f_\theta : \theta \in \Theta_h\}$, resolving ties by a fixed rule. By (44), misclassification implies an L^2-error of at least $c\kappa h^\alpha/2$. Consequently,

$$\sup_{f\in\mathcal{F}(\alpha,r)} \mathbb{E}_f\|\widehat{f} - f\|_2^2 \ge \frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge c\kappa^2 h^{2\alpha}.$$

Taking the infimum over estimators proves the proposition. □

Proof of Theorem 3.1. Choose r_* sufficiently small for Lemma 4.1 and Proposition 7.1. The upper bound then holds for every subclass $\mathcal{F}(\alpha, r)$, $0 < r \le r_*$. Proposition 8.2, with its fixed κ chosen according to r, gives the matching lower bound. □

Remark 8.3. The effective denominator is $2\alpha + d + 3$, rather than the denominator obtained by treating the forward map as an isotropic operator of a single differential order. In the lower bound there are $d - 1$ angular directions and a two-derivative attenuation. In the upper bound, transport integration reduces the integrated variance of a twice differentiated, smoothed white noise from order $n^{-1}h^{-d-4}$ to order $n^{-1}h^{-d-3}$. The small-radius assumption supplies uniform convexity of the pressure and uniform control of this transport geometry.

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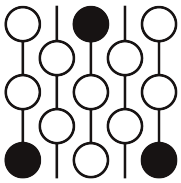
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Review of Problem 2 – Astra

First Proof Third Batch
Reviewer 3
October 8–9, 2026



1 Review

The manuscript proposes a kernel-based approach that first estimates the solution u of the underlying PDE and then reconstructs the unknown coefficient using the inversion formula in Eq. (14). The reconstruction is performed on *good events*, on which the plug-in estimator $\hat{u}$ satisfies suitable accuracy guarantees. The estimation risk is analysed using probabilistic arguments and PDE theory. The main result, Theorem 3.1, appears to be essentially correct: the stated convergence rates coincide with the expected minimax-optimal rates. The overall estimation procedure is intuitively plausible, and several key steps are motivated in a comprehensible manner. Although the statements presented may seem intuitively plausible, the corresponding proofs are frequently incomplete, heuristic, or mathematically not sufficiently explained. As a result, the validity of several claims cannot be reliably assessed from the proofs provided. Substantial revisions are therefore needed to bring the mathematical exposition and justification to the standard required for publication. Furthermore, the manuscript is insufficiently situated within the existing literature. It contains only four references, two of which are used to support well-known analytical facts in Section 4.1. That section at times devotes disproportionate attention to background material whose relevance to the main contribution remains unclear. A more focused presentation and a substantially broader discussion of related work would strengthen the manuscript.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: -0.5

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

Abstract

For a sufficiently small, fixed Hölder–Zygmund ball of conductivities around the constant function 1, the minimax squared L^2-risk is, up to logarithmic factors,

$$n^{-2\alpha/(2\alpha+d+3)}.$$

We prove matching polynomial upper and lower bounds for every $d \geq 2$ and $\alpha > 1$. The upper bound uses a regularized reconstruction along gradient trajectories, together with a variance estimate that exploits integration along these trajectories. The lower bound uses perturbations that oscillate in the $d-1$ angular directions. No boundary condition on the conductivity is imposed.

3 Statement and conventions

Let $D = B(0,1) \subset \mathbb{R}^d$. We interpret $C^s(D)$ as the Hölder–Zygmund restriction space $B^s_{\infty,\infty}(D)$. Any fixed usual equivalent norm may be used; the constants below, including the admissible radius, may depend on this choice.

White noise means the isonormal experiment

$$Y_n(\varphi) = \langle u_f, \varphi \rangle_{L^2(D)} + n^{-1/2}\mathbb{W}(\varphi), \qquad \varphi \in L^2(D),$$

where $\mathbb{E}[\mathbb{W}(\varphi)\mathbb{W}(\psi)] = \langle \varphi, \psi \rangle$. Write

$$\mathcal{F}(\alpha, r) = \{f : \|f - \mathbf{1}\|_{C^\alpha(D)} \leq r\}, \qquad R_n(\alpha, r) = \inf_{\widehat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\widehat{f} - f\|^2_{L^2(D)}.$$

Theorem 3.1. *Fix $d \geq 2$ and $\alpha > 1$. There exists $r_* > 0$ such that, for each fixed $r \in (0, r_*]$, there are constants $c, C > 0$ and n_0 for which*

$$c\, n^{-2\alpha/(2\alpha+d+3)} \leq R_n(\alpha, r) \leq C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)}, \qquad n \geq n_0.$$

If $\alpha \notin \mathbb{N}$, the logarithmic factor in this upper bound can be omitted. In particular, the answer up to logarithmic factors is

$$\boxed{R_n(\alpha, r) = \eqsim n^{-2\alpha/(2\alpha+d+3)}.}$$

For a smooth function a, the notation

$$\|a\|_{k,\infty;\Omega} = \max_{|\beta| \leq k} \|\partial^\beta a\|_{L^\infty(\Omega)}$$

denotes the classical derivative norm, not an integer-index Zygmund norm. Domain subscripts will be suppressed when unambiguous. Constants denoted by C may change from line to line, but are independent of n, the bandwidth, and the conductivity in the specified parameter class.

4 Analytic preliminaries

4.1 Function spaces and the forward equation

> **Reviewer 3**
>
> The detailed exposition of standard analytical tools, such as multiplication and interpolation inequalities, seems misplaced in this section, particularly given the insufficient care taken to introduce notation and terminology.

We use the Littlewood–Paley characterization

$$\|a\|_{C^s(\mathbb{R}^d)} \asymp \sup_{j\ge 0} 2^{js}\|\Delta_j a\|_\infty \tag{1}$$

for an inhomogeneous smooth dyadic decomposition. The equivalence with classical Hölder spaces at noninteger positive indices, and the differentiation properties of these spaces, are given in [3, Chapter 13, Section 8].

We also use a common bounded extension operator from a smooth bounded domain to $\mathbb{R}^d$, bounded on all the finitely many Besov spaces under consideration. Such an operator exists even for Lipschitz domains; see [2].

Here are some consequences, with the relevant arguments recalled. Differentiation maps C^s boundedly to C^{s-1}. For $s > 0$, C^s is a multiplication algebra:

$$\|ab\|_{C^s} \le C_s\|a\|_{C^s}\|b\|_{C^s}. \tag{2}$$

Indeed, decompose the product into low–high, high–low, and comparable-frequency products. The low–high contribution to its j-th dyadic block is bounded by $C\|a\|_\infty 2^{-js}\|b\|_{C^s}$; interchange a, b for the high–low contribution. For comparable frequencies $k \ge j - C$, the bound is

$$C\|a\|_{C^s}\|b\|_{C^s}\sum_{k\ge j-C} 2^{-2ks} \le C_s 2^{-js}\|a\|_{C^s}\|b\|_{C^s}.$$

This proves (2). Extension and restriction give the same facts on D.

We shall also use the real interpolation identity

$$(C^{s_0}, C^{s_1})_{\theta,\infty} = C^{(1-\theta)s_0+\theta s_1}, \qquad s_0 < s_1, \quad 0 < \theta < 1. \tag{3}$$

For completeness, on $\mathbb{R}^d$ this follows by splitting the dyadic series at a level J. If $s = (1-\theta)s_0 + \theta s_1$, the low and high parts satisfy

$$\|a_{\le J}\|_{C^{s_1}} \le C2^{J(s_1-s)}\|a\|_{C^s}, \qquad \|a_{>J}\|_{C^{s_0}} \le C2^{-J(s-s_0)}\|a\|_{C^s}.$$

Taking $2^{J(s_1-s_0)} \asymp t^{-1}$ bounds the K-functional by $Ct^\theta\|a\|_{C^s}$. Conversely, applying Δ_j to any decomposition $a = a_0 + a_1$ gives

$$\|\Delta_j a\|_\infty \le C2^{-js_0}K(2^{-j(s_1-s_0)}, a; C^{s_0}, C^{s_1}),$$

which proves the reverse bound. A common extension operator transfers the identity to D.

> **Reviewer 3**
>
> Although Lemma 4.1 is presented as self-contained and accompanied by a full proof with appropriate references, parts of the argument remain heuristic. Moreover, the result is not adequately situated within the existing statistical literature on PDE-based inverse problems. The same applies to the presentation of the proofs for Lemma 4.2 and Lemma 4.3 in the following section.

Lemma 4.1 (Uniform perturbative regularity)**.** *Put*

$$u_\circ(x) = \frac{|x|^2 - 1}{2d}.$$

For sufficiently small $r_* > 0$, every $f \in \mathcal{F}(\alpha, r_*)$ satisfies $1/2 \le f \le 3/2$, and the Dirichlet problem

$$\operatorname{div}(f\nabla u) = 1, \qquad u|_{\partial D} = 0$$

has a unique weak solution. Moreover,

$$\|u_f - u_\circ\|_{C^{\alpha+1}(D)} \le C\|f - \mathbf{1}\|_{C^\alpha(D)}. \tag{4}$$

Proof. Let $\mathcal{P}F$ be the zero-Dirichlet solution of $\Delta w = \operatorname{div} F$. For every $s > 1$,

$$\|\mathcal{P}F\|_{C^{s+1}(D)} \le C_s\|F\|_{C^s(D)}. \tag{5}$$

At noninteger s, this is the classical global Schauder estimate and solvability theorem, including higher-order regularity; see [1, Theorems 6.6, 6.14 and 6.19]. At integer s, apply these bounds at two surrounding noninteger indices and use (3). Thus (5) holds also at the Zygmund indices.

Writing $u = u_\circ + w$, the desired equation becomes

$$w = -\mathcal{P}\big((f-1)\nabla u_\circ\big) - \mathcal{P}\big((f-1)\nabla w\big).$$

By (2) and (5), the second term is a linear operator on the zero-trace subspace of $C^{\alpha+1}(D)$ with norm at most $C\|f-1\|_{C^\alpha}$. Choosing r_* so that this norm is at most $1/2$, the Neumann series gives a solution and (4).

The embedding $C^\alpha(D) \hookrightarrow C^0(\overline{D})$ allows us to decrease r_* so that $1/2 \le f \le 3/2$. Uniqueness of a weak solution follows by testing the equation for the difference of two solutions against that difference: $\int_D f|\nabla w|^2 = 0$. Finally, the constructed solution has zero trace and belongs to $H^1_0(D)$, so it is that unique weak solution. □

4.2 Smoothing and a product cancellation

Set

$$m_0 = \lceil \alpha \rceil - 1.$$

Fix a real $K \in C^\infty_c(B(0,1))$ such that

$$\int K = 1, \qquad \int x^\beta K(x)\,dx = 0 \quad (1 \le |\beta| \le 2m_0 + 2). \tag{6}$$

Such a kernel can be obtained from a finite linear combination of distinct dilations of a smooth unit-integral bump: the required coefficients solve a Vandermonde system in the dilation parameters. Positivity of K is neither required nor asserted.

For $0 < h < 1/8$, put

$$K_h(x) = h^{-d}K(x/h), \qquad \Omega_h = B(0, 1-2h), \qquad S_h a(x) = \int_D K_h(x-y)a(y)\,dy, \quad x \in \Omega_h,$$

and let $L_h = \log(e/h)$.

Lemma 4.2 (Smoothing and commutator bounds)**.** *Suppose $a, b \in C^\alpha(D)$ have bounded norms. Uniformly on Ω_h,*

$$\|S_h a - a\|_\infty \le C h^\alpha L_h, \tag{7}$$

$$\big\|\partial^\gamma\big((S_h a)(S_h b) - S_h(ab)\big)\big\|_\infty \le C h^{\alpha+1-|\gamma|} L_h^2, \qquad |\gamma| \le 2. \tag{8}$$

If $\alpha \notin \mathbb{N}$, all factors L_h in these two bounds may be replaced by 1. The statements hold componentwise for vector-valued b.

Proof. We first record the local Taylor remainder

$$\sup_{|y-x|\le h} |a(y) - P_{a,x}(y)| \le C h^\alpha L_h, \tag{9}$$

where $P_{a,x}$ is the Taylor polynomial of degree m_0. Its coefficients are uniformly bounded. At noninteger α, this is the ordinary Hölder Taylor estimate, without L_h. At integer α, derivatives of order $\alpha - 1$ have modulus of continuity bounded by $Ct\log(e/t)$. To see the latter directly from (1), split their dyadic series at $2^J \asymp t^{-1}$: the low-frequency increments sum to $C\sum_{j\le J} t$, and the high-frequency tail is at most $C\sum_{j>J} 2^{-j}$. Taylor's integral remainder then gives (9). Extension to $\mathbb{R}^d$ justifies this argument uniformly near the boundary.

Polynomial reproduction from (6) immediately proves (7). To prove (8), write

$$Q_h(a, b) = (S_h a)(S_h b) - S_h(ab).$$

Fix x and write $a = P_a + R_a$, $b = P_b + R_b$, using the Taylor polynomials at x. Keep these polynomials fixed when differentiating at x. Polynomial reproduction gives $Q_h(P_a, P_b) = 0$. Also $Q_h(a, c) = 0$ for every constant c.

On $B(x, h)$, the remainders satisfy $|R_a| + |R_b| \le C h^\alpha L_h$, whereas $|P_a - P_a(x)| + |P_b - P_b(x)| \le Ch$. Differentiating the convolution kernels gives, for $|\gamma| \le 2$,

$$|\partial^\gamma Q_h(R_a, P_b)(x)| + |\partial^\gamma Q_h(P_a, R_b)(x)| \le C h^{\alpha+1-|\gamma|} L_h.$$

For example, in the first expression subtract the constant $P_b(x)$; the differentiated convolution of $R_a(P_b - P_b(x))$ has precisely this bound. In the product term, a derivative falling on the polynomial is bounded by Ch^{1-k} at derivative order $k \ge 1$; the undifferentiated polynomial vanishes at x. Finally,

$$|\partial^\gamma Q_h(R_a, R_b)(x)| \le C h^{2\alpha-|\gamma|} L_h^2.$$

Since $\alpha > 1$, these inequalities prove (8). The same proof without logarithms applies at noninteger α. □

Lemma 4.3 (Completion of a thin boundary strip)**.** *For all sufficiently small h, there is a linear map $E_h : L^2(\Omega_h) \to L^2(D)$ such that*

$$\|E_h a\|_{L^2(D)} \le C \|a\|_{L^2(\Omega_h)}, \tag{10}$$

$$\|E_h(f|_{\Omega_h}) - f\|_{L^2(D)} \le C \|f\|_{C^\alpha(D)} h^\alpha L_h. \tag{11}$$

At noninteger α, L_h can be omitted.

Proof. Let $q = m_0 + 1$ and $\lambda_j = (-1)^{j+1}\binom{q}{j}$, $1 \le j \le q$. Then

$$\sum_{j=1}^{q} \lambda_j = 1, \qquad \sum_{j=1}^{q} \lambda_j j^k = 0, \quad 1 \le k \le m_0.$$

Define $E_h a = a$ on Ω_h, and, for $x \in D \setminus \Omega_h$, define

$$E_h a(x) = \sum_{j=1}^{q} \lambda_j a((1 - 3jh)x).$$

For $h < 1/(6q)$, all arguments on the right belong to Ω_h and all dilation factors are at least $1/2$. A change of variables proves (10). Taylor expansion of f at x, together with the displayed moment identities and (9), proves (11). □

5 Inversion along gradient trajectories

For a smooth function v, define

$$T_v a = \operatorname{div}(a \nabla v) = \nabla v \cdot \nabla a + (\Delta v) a.$$

Lemma 5.1 (Transport inverse)**.** *Let $\Omega = B(0, R)$, $3/4 \le R \le 1$, and suppose v is smooth on a neighborhood of $\overline{\Omega}$, with*

$$\lambda I \le D^2 v \le \Lambda I, \qquad \nabla v \cdot \nu > 0 \text{ on } \partial\Omega, \tag{12}$$

where $0 < \lambda \le \Lambda$. There is a unique critical point $p \in \Omega$. Let $z_t(x)$ solve

$$\frac{d}{dt} z_t(x) = -\nabla v(z_t(x)), \qquad z_0(x) = x.$$

This flow exists in $\overline{\Omega}$ for all $t \ge 0$, and

$$e^{-\Lambda t}|x - p| \le |z_t(x) - p| \le e^{-\lambda t}|x - p|. \tag{13}$$

For smooth g, the unique bounded smooth solution of $T_v a = g$ is

$$P_v g(x) = \int_0^\infty \exp\left\{ -\int_0^t \Delta v(z_s(x))\, ds \right\} g(z_t(x))\, dt. \tag{14}$$

With $c_0 = d\lambda$, it satisfies

$$\|P_v g\|_\infty \le c_0^{-1} \|g\|_\infty, \tag{15}$$

$$\|P_v g\|_2 \le 2 c_0^{-1} \|g\|_2, \tag{16}$$

$$\|P_v g\|_{H^1} \le C\big\{ \|g\|_{H^1} + (1 + \|D^3 v\|_\infty)\|g\|_2 \big\}. \tag{17}$$

The constants in the last estimate depend only on d, λ, Λ.

Proof. Strict convexity gives at most one critical point. A minimum on $\overline{\Omega}$ cannot occur at the boundary, because the inward direction decreases v, so this critical point exists in Ω. The vector field $-\nabla v$ points inward at the boundary; the usual first-exit argument and local existence for a smooth vector field therefore give a global backward-gradient flow in $\overline{\Omega}$.

For $z \in \Omega$,

$$\lambda|z-p|^2 \le (z-p)\cdot \nabla v(z) \le \Lambda |z-p|^2.$$

Applying this to the derivative of $|z_t - p|$ proves (13).

Since $\Delta v \ge c_0$, the integral in (14) converges absolutely and gives (15). It is smooth up to the boundary. Here is a justification of differentiating the infinite-time integral. The first variational equation gives $\|D_x z_t\| \le e^{-\lambda t}$. Inductively, the equation for $D_x^k z_t$ has homogeneous part $-D^2 v(z_t) D_x^k z_t$, while each forcing term is a product of at least two lower-order flow derivatives. Variation of constants therefore gives $\|D_x^k z_t\| \le C_k e^{-\lambda t}$. Derivatives of $\int_0^t \Delta v(z_s)\, ds$ of every positive order are consequently bounded uniformly in t. Every differentiated integrand is thus bounded by $C_k e^{-c_0 t}$.

The flow semigroup property, or differentiation after splitting the integral at a finite time, gives $T_v P_v g = g$. If $T_v a = 0$, differentiation along the flow shows

$$a(x) = \exp\left\{-\int_0^t \Delta v(z_s(x))\, ds\right\} a(z_t(x)).$$

Letting $t \to \infty$ proves uniqueness among bounded solutions.

For $a = P_v g$, integration by parts gives

$$\int_\Omega a T_v a = \frac{1}{2}\int_{\partial\Omega} (\nabla v \cdot \nu) a^2 + \frac{1}{2}\int_\Omega (\Delta v) a^2 \ge \frac{c_0}{2}\|a\|_2^2.$$

This proves (16). Differentiating the equation gives

$$T_v(\nabla a) + D^2 v\, \nabla a = \nabla g - a \nabla \Delta v.$$

The same integration by parts, now for a vector field, yields

$$(c_0/2 + \lambda)\|\nabla a\|_2 \le \|\nabla g\|_2 + \|\nabla \Delta v\|_\infty \|a\|_2.$$

Together with (16), this proves (17). □

6 Gaussian smoothing and the decisive variance estimate

Define

$$Z_h(x) = n^{-1/2}\mathbb{W}(K_h(x-\cdot)), \qquad x \in \mathbb{R}^d,$$

where the white-noise argument is restricted to D.

Lemma 6.1 (Gaussian smoothing bounds)**.** *For every fixed nonnegative integer k and $p \ge 1$,*

$$\mathbb{E}\|Z_h\|_{k,\infty;\Omega_h}^p \le C_p n^{-p/2} h^{-p(d/2+k)} L_h^{p/2}, \tag{18}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^2 \le C n^{-1} h^{-d-2k}, \tag{19}$$

$$\mathbb{E}\|Z_h\|_{H^k(\Omega_h)}^4 \le C n^{-2} h^{-2d-4k}. \tag{20}$$

Moreover, for $t \ge 0$,

$$\mathbb{P}\left(\|Z_h\|_{k,\infty;\Omega_h} > C n^{-1/2} h^{-d/2-k}\sqrt{L_h + t}\right) \le C e^{-t}. \tag{21}$$

Proof. The white-noise isometry gives

$$\mathbb{E}|\partial^\beta Z_h(x)|^2 \le Cn^{-1}h^{-d-2|\beta|}.$$

Integrating proves (19). For centered jointly Gaussian variables X, Y, $\mathbb{E}(X^2Y^2) \le 3(\mathbb{E}X^2)(\mathbb{E}Y^2)$; double integration proves (20).

The field has a smooth version. Indeed, after multiplication by a fixed smooth cutoff, its weak derivatives of every order have finite expected squared L^2-norm. Almost surely it therefore belongs to every integer Sobolev space locally. Fourier inversion and Cauchy–Schwarz imply $H^s(\mathbb{R}^d) \hookrightarrow C^k$ for $s > k + d/2$, which gives the assertion.

For a derivative of order k, let $A = n^{-1/2}h^{-d/2-k}$. Its Gaussian increment metric is bounded by

$$CA\min\{1, |x-y|/h\}.$$

On a fixed cube containing Ω_h, use successive grids of mesh $h2^{-j}$. The initial grid has at most Ch^{-d} points, and the j-th grid has at most $Ch^{-d}2^{jd}$ points. Join each point to a nearest point of the preceding grid. Gaussian tail bounds and a union bound show that, outside an event of probability Ce^{-t}, the initial values are bounded by $CA\sqrt{L_h + t}$, and every increment at level $j \ge 1$ is bounded by

$$CA2^{-j}\sqrt{L_h + t + j}.$$

The constants can be chosen so that the failure probabilities at level j sum to Ce^{-t}. Summing the increments and using continuity proves (21). A finite union handles all derivatives of order at most k. Integrating the tail proves (18).

> **Reviewer 3**
>
> Establishing the concentration inequality in Eq. (21) is in general nontrivial and is a key step in the proof. Although a heuristic justification is provided, the argument lacks the precision and detail needed for rigorous verification. In particular, the use successive grids is insufficiently specified, leaving a substantial gap in the proof.

□

The next lemma is responsible for the denominator $2\alpha + d + 3$.

Lemma 6.2 (Variance after transport inversion)**.** *Suppose v satisfies* (12) *on Ω_h, and*

$$2d\lambda > \Lambda.$$

If $\|a\|_{1,\infty;\Omega_h} \le A_0$, then

$$\mathbb{E}\,\|P_v \operatorname{div}(a\nabla Z_h)\|^2_{L^2(\Omega_h)} \le Cn^{-1}h^{-d-3}. \tag{22}$$

The constant depends only on $d, \lambda, \Lambda, A_0, K$, and not on higher derivatives of v or a.

Proof. For $z \in \Omega_h$, set

$$H_h(z,y) = a(z)\Delta K_h(z-y) + \nabla a(z)\cdot\nabla K_h(z-y).$$

Its $L^2(D)$-norm is at most $Ch^{-d/2-2}$, and its support in y is contained in $B(z,h)$. Consequently,

$$|\langle H_h(z,\cdot), H_h(w,\cdot)\rangle| \le Ch^{-d-4}\mathbf{1}_{\{|z-w|\le 2h\}}. \tag{23}$$

Let p be the critical point of v, $c_0 = d\lambda$, and $r = |x - p|$. The representation (14), the white-noise isometry, and (23) give the pointwise variance bound

$$Cn^{-1}h^{-d-4}\int_0^\infty \int_0^\infty e^{-c_0(t+s)}\mathbf{1}_{\{|z_t(x)-z_s(x)|\le 2h\}}\,ds\,dt.$$

For $s = t + w$, put $r_t = |z_t(x) - p|$. By (13),

$$|z_t - z_{t+w}| \ge r_t - r_{t+w} \ge (1 - e^{-\lambda w})r_t.$$

If $r_t > 4h$, overlap therefore forces $w \le 4h/(\lambda r_t)$. If $r_t \le 4h$, the exponential integral in w is bounded by a constant. Thus

$$\int_0^\infty e^{-c_0 w}\mathbf{1}_{\{|z_t-z_{t+w}|\le 2h\}}\,dw \le C\min\{1, h/r_t\}.$$

Since $r_t \ge e^{-\Lambda t}r$, the double time integral is at most

$$C\int_0^\infty e^{-2c_0 t}\min\{1, (h/r)e^{\Lambda t}\}\,dt \le C\min\{1, h/r\}.$$

The last inequality uses $2c_0 > \Lambda$; at $r = 0$ the minimum is interpreted as 1.

Finally, since $d \ge 2$ and $\Omega_h \subset B(p, 2)$,

$$\int_{\Omega_h}\min\{1, h/|x-p|\}\,dx \le C\left(h^d + h\int_h^2 r^{d-2}\,dr\right) \le Ch.$$

Integrating the pointwise variance proves (22). All interchanges of integrals are justified by the exponential weights and the uniform L^2-bound on H_h. □

> **Reviewer 3**
>
> The statement of Lemma 6.2 is a key argument in deriving the correct rate of convergence. The proof employs intuitively sound arguments – in particular, it involves an explicit examination of the critical point v. Nevertheless, the arguments presented are also rather heuristic and, in some cases, insufficiently explained; for instance, the w in Eq. 23 and the x in the line following it are not specified, which makes it possible to follow the precise arguments only with considerable effort.

7 The upper bound

Proposition 7.1. *For sufficiently small $r_* > 0$, the upper bound in Theorem 3.1 holds uniformly over $\mathcal{F}(\alpha, r_*)$.*

Proof. Write

$$p_\alpha = 2\alpha + d + 3, \qquad h = n^{-1/p_\alpha}.$$

Only sufficiently large n will be considered.

Construction of the estimator

Form the observable smooth field

$$\widehat{u}_h(x) = Y_n(K_h(x - \cdot)), \qquad x \in \Omega_h.$$

For a fixed true f, use the deterministic notation

$$u = u_f, \qquad v = S_h u, \qquad a = S_h f.$$

Then $\widehat{u}_h = v + Z_h$.

Choose a fixed $\delta > 0$, depending only on d, so small that

$$\|w - u_\circ\|_{2,\infty;\Omega_h} \le \delta$$

> **Reviewer 3**
>
> What is w? Is this actually meant to be v?

implies (12), uniformly for $h < 1/8$, with

$$\lambda = \frac{3}{4d}, \qquad \Lambda = \frac{5}{4d}.$$

Indeed, sufficiently small entrywise perturbations of $D^2 u_\circ = I/d$ give these Hessian bounds, and $\nabla u_\circ \cdot \nu = (1-2h)/d$ on $\partial\Omega_h$. These constants also satisfy $2d\lambda > \Lambda$.

Polynomial reproduction and Lemma 4.1 imply

$$\|v - u_\circ\|_{2,\infty} \le C r_*.$$

Also, by moving one derivative onto the kernel,

$$\|a\|_{1,\infty} \le C, \qquad \|a\|_{2,\infty} \le C h^{-1}, \qquad \|v\|_{3,\infty} \le C h^{-1}.$$

> **Reviewer 3**
>
> The use of a varies throughout the manuscript from a general function (see Lemma 4.2, Lemma 4.3) to functions a that are implicitly defined by $T_v a$ (in the proof of Lemma 5.1). Its use here in the proof of Proposition 7.1 is therefore unclear.

Decrease r_* so that the first bound is at most $\delta/2$. Choose a fixed sufficiently large M_0, and define the observable event

$$\mathcal{A}_h = \left\{ \|\widehat{u}_h - u_\circ\|_{2,\infty} \le \delta, \quad h\|\widehat{u}_h\|_{3,\infty} \le M_0 \right\}.$$

Lemma 6.1 implies, uniformly in the parameter class,

$$\mathbb{P}_f(\mathcal{A}_h^c) \le C n^{-3}. \tag{24}$$

To verify this explicitly, take $t = 3\log n$ in (21). Both the C^2-noise norm and h times the C^3-noise norm are bounded, with this probability, by a constant times

$$n^{-1/2} h^{-d/2-2} \sqrt{\log n} = h^{\alpha - 1/2} \sqrt{\log n} = o(1).$$

On $\mathcal{A}_h$, define

$$\widetilde{f}_h = P_{\widehat{u}_h} \mathbf{1} \quad \text{on } \Omega_h.$$

Let Π denote pointwise projection onto $[1/2, 3/2]$, and set

> **Reviewer 3**
>
> There is no justification for limiting the range of values exclusively to between 1/2 and 3/2, nor is it clear whether this is sufficient or necessary, or whether general restrictions are possible.

$$\widehat{f}_n = \begin{cases} \Pi(E_h \widetilde{f}_h), & \mathcal{A}_h, \\ \mathbf{1}, & \mathcal{A}_h^c. \end{cases}$$

This is an estimator based only on the observations. Its measurability follows from the flow formula: finite-time solutions of the smooth ODE depend continuously on the vector field, and the tail of (14) is uniformly exponentially small on $\mathcal{A}_h$. Notice that the auxiliary transport reconstruction does not require a boundary value for $\widetilde{f}_h$.

Deterministic smoothing bias

Set

$$\mathcal{R}_h = 1 - T_v a.$$

Because $\operatorname{div}(f\nabla u) = 1$, convolution commutes with differentiation in the interior and gives

$$\mathcal{R}_h = -\operatorname{div}\left((S_h f)(S_h \nabla u) - S_h(f\nabla u)\right).$$

Both f and ∇u have uniformly bounded C^α-norms. Lemma 4.2 therefore gives, with

$$B_h = h^\alpha L_h^2,$$

the bounds

$$\|\mathcal{R}_h\|_2 \le C B_h, \qquad \|\mathcal{R}_h\|_{H^1} \le C h^{-1} B_h. \tag{25}$$

Also, Lemmas 4.2 and 4.3 give

$$\|E_h a - f\|_{L^2(D)} \le C h^\alpha L_h. \tag{26}$$

Control of the nonlinear noise contribution

All identities in this paragraph are on $\mathcal{A}_h$. Put

$$e = \widetilde{f}_h - a, \qquad S_Z b = \operatorname{div}(b \nabla Z_h).$$

Since $T_{\widehat{u}_h} = T_v + S_Z$,

$$T_{\widehat{u}_h} e = \mathcal{R}_h - S_Z a. \tag{27}$$

The elementary product bounds

$$\|S_Z a\|_2 \le C\|Z_h\|_{H^2}, \qquad \|S_Z a\|_{H^1} \le C\big(\|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big)$$

follow by differentiating $a\Delta Z_h + \nabla a \cdot \nabla Z_h$ and using the displayed bounds on a. Lemma 5.1, the definition of $\mathcal{A}_h$, and (25) yield the rough estimates

$$\|e\|_2 \le C\big(B_h + \|Z_h\|_{H^2}\big), \tag{28}$$

$$\|e\|_{H^1} \le C\big(h^{-1}B_h + \|Z_h\|_{H^3} + h^{-1}\|Z_h\|_{H^2}\big). \tag{29}$$

To obtain the sharp variance, rewrite (27) as

$$e = P_v \mathcal{R}_h - P_v S_Z a - P_v S_Z e. \tag{30}$$

The middle term is linear in the noise and, by Lemma 6.2,

$$\mathbb{E}\|P_v S_Z a\|_2^2 \le C n^{-1} h^{-d-3}. \tag{31}$$

The last term is controlled without any independence assumption between the estimated flow and the noise. Namely,

$$\|P_v S_Z e\|_2 \le C\left(\|Z_h\|_{1,\infty}\|e\|_{H^1} + \|Z_h\|_{2,\infty}\|e\|_2\right).$$

Substitute (28)–(29) and apply Lemma 6.1 and Cauchy–Schwarz. This gives

$$\mathbb{E}\left[\mathbf{1}_{\mathcal{A}_h}\|P_v S_Z e\|_2^2\right] \le C\left(B_h^2 n^{-1} h^{-d-4} L_h + n^{-2} h^{-2d-8} L_h\right). \tag{32}$$

For clarity, a representative fourth-moment calculation is

$$\mathbb{E}\left(\|Z_h\|_{1,\infty}^2 \|Z_h\|_{H^3}^2\right) \le \left(\mathbb{E}\|Z_h\|_{1,\infty}^4\right)^{1/2} \left(\mathbb{E}\|Z_h\|_{H^3}^4\right)^{1/2} \le C n^{-2} h^{-2d-8} L_h.$$

The term with $h^{-1}\|Z_h\|_{1,\infty}\|Z_h\|_{H^2}$ has the same bound, as does the term with $\|Z_h\|_{2,\infty}\|Z_h\|_{H^2}$. The terms containing B_h give the first term in (32).

Combining (30)–(32),

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}_h}\|e\|_2^2] \le C\left(B_h^2 + n^{-1}h^{-d-3} + B_h^2 n^{-1} h^{-d-4} L_h + n^{-2} h^{-2d-8} L_h\right).$$

Our bandwidth satisfies $n^{-1} = h^{2\alpha+d+3}$. Hence the last display becomes

$$C\left(h^{2\alpha} L_h^4 + h^{2\alpha} + h^{2\alpha} L_h^4 h^{2\alpha-1} L_h + h^{4\alpha-2} L_h\right) \le C h^{2\alpha} L_h^4.$$

Here $\alpha > 1$ is used in $h^{4\alpha-2} L_h \le C h^{2\alpha}$.

Projection onto $[1/2, 3/2]$ cannot increase the loss, since the true conductivity takes values in that interval. By (10), (26), and (24),

$$\sup_{f \in \mathcal{F}(\alpha, r_*)} \mathbb{E}_f \|\widehat{f}_n - f\|_2^2 \le C h^{2\alpha} L_h^4 + C n^{-3}.$$

This is the asserted upper bound.

If $\alpha \notin \mathbb{N}$, use the versions of the approximation lemmas without logarithms, so that $B_h = h^\alpha$. The logarithms remaining in the stochastic remainder are absorbed by the strictly positive powers $h^{2\alpha-1}$ and $h^{2\alpha-2}$. The resulting upper bound is $Ch^{2\alpha}$. □

8 The lower bound

> **Reviewer 3**
>
> The proof of the lower bound follows the general framework described in detail in [4], which is only hinted at in the reference to Fano's inequality in the proof of Proposition 8.2. To meet scientific standards, a comprehensive literature review is essential, and one should clearly situate this at the beginning of Section 8. Furthermore, various notations and mathematical arguments introduced and presented below - such as Q, $\mathbb{S}^{d-1}$, and $\mathcal{H}_k$ - are introduced and presented only heuristically, which makes it difficult to verify them completely.

We construct pressure perturbations with fine angular oscillations and fixed radial scale.

8.1 Angular bumps

Let $m = d-1$. Fix a smooth coordinate chart $\chi : Q \to \mathbb{S}^{d-1}$, and a smaller cube compactly contained in Q. For sufficiently small h, choose centers $z_1, \dots, z_{M_h}$ in this cube, separated by $4h$, such that

$$ch^{-m} \le M_h \le Ch^{-m}.$$

Fix a nonzero $\psi \in C_c^\infty((-1,1)^m)$, and define

$$\psi_{j,h}(\chi(z)) = \psi((z - z_j)/h),$$

extended by zero outside the chart. Their supports are pairwise disjoint. Coordinate changes on the fixed chart give

$$\|\psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^m, \tag{33}$$

$$\|\nabla_{\mathbb{S}^{d-1}} \psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^{m-2}. \tag{34}$$

Integration by parts and Cauchy–Schwarz imply

$$\|\Delta_{\mathbb{S}^{d-1}} \psi_{j,h}\|_2^2 \ge \frac{\|\nabla_{\mathbb{S}^{d-1}} \psi_{j,h}\|_2^4}{\|\psi_{j,h}\|_2^2} \ge ch^{m-4}. \tag{35}$$

For $\theta \in \{0,1\}^{M_h}$, write $q_\theta = \sum_j \theta_j \psi_{j,h}$. Disjointness gives

$$\|q_\theta\|_2^2 \le C, \qquad \|\nabla^k_{\mathbb{S}^{d-1}} q_\theta\|_\infty \le C_k h^{-k}$$

for every fixed k.

8.2 Realization as conductivities

Fix a nonnegative, nonzero $\rho \in C_c^\infty((1/4, 1/2))$. For a fixed small $\kappa > 0$, define on $\mathbb{R}^d$

$$v_\theta(x) = \kappa h^{\alpha+2} \rho(|x|) q_\theta\left(\frac{x}{|x|}\right), \qquad u_\theta = u_\circ + v_\theta.$$

The perturbation is zero near 0, so this defines a smooth function everywhere. For each fixed k,

$$\|\nabla^k v_\theta\|_\infty \le C_k \kappa h^{\alpha+2-k}. \tag{36}$$

In particular, for sufficiently small κ, $D^2 u_\theta$ is uniformly positive definite, independently of h, θ.

Lemma 8.1 (Conductivity realization and expansion)**.** *For sufficiently small κ, there are smooth positive conductivities $f_\theta = 1 + g_\theta$ such that*

$$\operatorname{div}(f_\theta \nabla u_\theta) = 1 \quad \text{on } \mathbb{R}^d,$$

and, uniformly in h, θ,

$$\|\nabla^k g_\theta\|_\infty \le C_k \kappa h^{\alpha-k} \quad \text{for every fixed } k, \tag{37}$$

$$\|g_\theta\|_{C^\alpha(D)} \le C\kappa, \tag{38}$$

$$g_\theta = \ell_\theta + \varepsilon_\theta, \qquad \ell_\theta = -P_{u_\circ} \Delta v_\theta, \qquad \|\varepsilon_\theta\|_{L^2(D)} \le C\kappa^2 h^{2\alpha}. \tag{39}$$

Moreover, $u_{f_\theta} = u_\theta$ on D.

Proof. The backward-gradient flow of u_θ is globally defined on $\mathbb{R}^d$: its vector field is globally Lipschitz and has at most linear growth. Its unique critical point is 0, and the flow contracts towards 0. Define $f_\theta = P_{u_\theta}\mathbf{1}$ by the same integral as (14). That formula and its proof apply on $\mathbb{R}^d$ here, since the Hessian has uniform positive bounds and all derivatives of order at least two are bounded. Thus f_θ is smooth, positive, bounded, and satisfies the stated equation.

Write $u = u_\theta$, $v = v_\theta$, and $g = f_\theta - 1$. Then

$$T_u g = -\Delta v. \tag{40}$$

The L^∞ transport estimate gives $\|g\|_\infty \le C\kappa h^\alpha$.

We prove (37) by induction. Differentiating (40) k times, and collecting the terms containing k-th derivatives of g, gives

$$T_u(\nabla^k g) + \mathcal{H}_k \nabla^k g = F_k.$$

Here $\mathcal{H}_k$ is the sum of the actions of D^2u on the k tensor indices. If $D^2u \ge \lambda I$, it is symmetric and bounded below by $k\lambda I$. Leibniz's rule gives

$$\begin{aligned}\|F_k\|_\infty \le C_k\Bigg(&\|\nabla^{k+2} v\|_\infty \\ &+ \sum_{j=2}^{k} \|\nabla^{j+1} v\|_\infty \|\nabla^{k-j+1} g\|_\infty \\ &+ \sum_{j=1}^{k} \|\nabla^{j+2} v\|_\infty \|\nabla^{k-j} g\|_\infty \Bigg).\end{aligned}$$

All derivatives of $u_\circ$ of order at least three vanish. Along each trajectory, the homogeneous matrix equation has propagator norm at most $e^{-(d+k)\lambda t}$. Consequently

$$\|\nabla^k g\|_\infty \le ((d+k)\lambda)^{-1} \|F_k\|_\infty.$$

The differentiability and boundedness needed here also follow by the variational-equation argument in Lemma 5.1. Using the induction hypothesis and (36), the right side is bounded by

$$C_k\big(\kappa h^{\alpha-k} + \kappa^2 h^{2\alpha-k}\big) \le C_k' \kappa h^{\alpha-k}.$$

This proves (37).

Choose an integer $J > \alpha$. The dyadic characterization implies

$$\|g\|_{C^\alpha(\mathbb{R}^d)} \le C\left(\|g\|_\infty + \|g\|_\infty^{1-\alpha/J} \|\nabla^J g\|_\infty^{\alpha/J}\right).$$

Indeed, a dyadic block is bounded both by $C\|g\|_\infty$ and, using its vanishing moments, by $C2^{-jJ}\|\nabla^J g\|_\infty$; optimize between these bounds. Substitution of (37) proves (38), including at integer α.

Expanding the original equation around $(1, u_\circ)$ gives

$$T_{u_\circ} g = -\Delta v - \operatorname{div}(g\nabla v).$$

Therefore

$$g = -P_{u_\circ}\Delta v - P_{u_\circ}\operatorname{div}(g\nabla v).$$

The L^∞ bound for $P_{u_\circ}$, together with (36) and (37) at orders zero and one, yields

$$\|P_{u_\circ}\operatorname{div}(g\nabla v)\|_\infty \le C\left(\|\nabla g\|_\infty \|\nabla v\|_\infty + \|g\|_\infty \|\Delta v\|_\infty\right) \le C\kappa^2 h^{2\alpha}.$$

This proves (39).

Finally, v_θ vanishes near ∂D, so $u_\theta = 0$ there. Since $f_\theta > 0$, uniqueness of the Dirichlet problem implies $u_{f_\theta} = u_\theta$. □

8.3 Separation of the conductivities

For $x = r\omega$, the unperturbed inverse is explicitly

$$P_{u_\circ} s(r\omega) = d r^{-d} \int_0^r t^{d-1} s(t\omega)\, dt.$$

Using the polar-coordinate Laplacian, for $r > 1/2$ we obtain

$$\ell_\theta(r\omega) = -d\kappa h^{\alpha+2} J_\rho r^{-d} \Delta_{\mathbb{S}^{d-1}} q_\theta(\omega), \qquad J_\rho = \int_0^\infty t^{d-3} \rho(t)\, dt > 0. \tag{41}$$

Indeed, the radial part integrates to $[t^{d-1}\rho'(t)]_0^r = 0$.

Let $H(\theta, \eta)$ be Hamming distance. Since the angular supports are disjoint, integration over the fixed annulus $3/4 < |x| < 7/8$, using (35) and (41), gives

$$\|\ell_\theta - \ell_\eta\|^2_{L^2(D)} \geq c\kappa^2 h^{2\alpha+m} H(\theta, \eta). \tag{42}$$

There is a subset $\Theta_h \subset \{0,1\}^{M_h}$ such that

$$\log |\Theta_h| \geq c M_h, \qquad H(\theta, \eta) \geq M_h/8 \quad (\theta \neq \eta). \tag{43}$$

This is the usual binary packing bound; see, for example, [4, Lemma 2.9]. One elementary proof is to select words greedily, removing at each step a Hamming ball of radius $\lfloor M_h/8 \rfloor$. Such a ball has at most

$$\sum_{k \leq M_h/8} \binom{M_h}{k} \leq \exp\{M_h H_b(1/8)\}$$

points, where $H_b(t) = -t\log t - (1-t)\log(1-t) < \log 2$. The binomial bound follows by multiplying by $(t/(1-t))^k$ and applying the binomial theorem. The greedy procedure therefore leaves exponentially many selected words and proves (43).

Since $M_h \asymp h^{-m}$, equations (42) and (43) imply

$$\|\ell_\theta - \ell_\eta\|_2 \geq c\kappa h^\alpha \quad (\theta, \eta \in \Theta_h,\ \theta \neq \eta).$$

Lemma 8.1 then gives, for all sufficiently small h,

$$\|f_\theta - f_\eta\|_2 \geq c'\kappa h^\alpha. \tag{44}$$

The remainder is negligible because $\kappa^2 h^{2\alpha} = o(\kappa h^\alpha)$.

8.4 Testing bound

Proposition 8.2. *For every fixed sufficiently small $r > 0$,*

$$R_n(\alpha, r) \geq c\, n^{-2\alpha/(2\alpha+d+3)}$$

for all sufficiently large n.

Proof. Choose a fixed positive κ so small that all the preceding constructions apply and $C\kappa \leq r$, where C is the constant in (38). Thus every f_θ belongs to $\mathcal{F}(\alpha, r)$.

Let P_θ be the observation law under f_θ, and let $P_\circ$ be that under $f = 1$. For Gaussian white noise with variance level n^{-1},

$$\mathrm{KL}(P_\theta, P_\circ) = \frac{n}{2}\|u_\theta - u_\circ\|_2^2 = \frac{n}{2}\|v_\theta\|_2^2 \le Cn\kappa^2 h^{2\alpha+4}. \tag{45}$$

The equality follows directly from the Gaussian likelihood

$$\frac{dP_\theta}{dP_\circ} = \exp\left\{n\big(Y_n(v_\theta) - \langle u_\circ, v_\theta\rangle\big) - \frac{n}{2}\|v_\theta\|_2^2\right\}.$$

It can equivalently be obtained by finite-dimensional projections and convergence of the isonormal series. The inequality in (45) uses $\|q_\theta\|_2^2 \le C$.

Take

$$h = n^{-1/(2\alpha+d+3)}.$$

Since $m = d - 1$,

$$nh^{2\alpha+4} = h^{-m}, \qquad \log|\Theta_h| \ge ch^{-m}.$$

Decrease the fixed κ, if necessary, so that

$$\frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathrm{KL}(P_\theta, P_\circ) \le \frac{1}{8}\log|\Theta_h|. \tag{46}$$

We recall the finite-hypothesis information inequality used here, a form of Fano's inequality; compare [4, Theorem 2.5 and Chapter 2]. Let J be uniform over N hypotheses and let $\overline{P} = N^{-1}\sum_j P_j$. For every reference law P_0,

$$I(J;Y) = \frac{1}{N}\sum_j \mathrm{KL}(P_j, \overline{P}) \le \frac{1}{N}\sum_j \mathrm{KL}(P_j, P_0),$$

because the difference of the right side and the middle expression is $\mathrm{KL}(\overline{P}, P_0)$. For any classifier $\widehat{J}$, with error probability p_e, the binary error indicator gives

$$H(J \mid Y) \le \log 2 + p_e \log N.$$

Since $H(J) = \log N$, it follows that

$$p_e \ge 1 - \frac{I(J;Y) + \log 2}{\log N}.$$

Applying this with (46) shows that every classifier has average error probability at least $1/2$ for large n.

Given any $L^2(D)$-valued estimator, classify by its nearest member of $\{f_\theta : \theta \in \Theta_h\}$, resolving ties by a fixed rule. By (44), misclassification implies an L^2-error of at least $c\kappa h^\alpha/2$. Consequently,

$$\sup_{f\in\mathcal{F}(\alpha,r)} \mathbb{E}_f\|\widehat{f} - f\|_2^2 \ge \frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge c\kappa^2 h^{2\alpha}.$$

Taking the infimum over estimators proves the proposition. □

Proof of Theorem 3.1. Choose r_* sufficiently small for Lemma 4.1 and Proposition 7.1. The upper bound then holds for every subclass $\mathcal{F}(\alpha, r)$, $0 < r \le r_*$. Proposition 8.2, with its fixed κ chosen according to r, gives the matching lower bound. □

Remark 8.3. The effective denominator is $2\alpha + d + 3$, rather than the denominator obtained by treating the forward map as an isotropic operator of a single differential order. In the lower bound there are $d - 1$ angular directions and a two-derivative attenuation. In the upper bound, transport integration reduces the integrated variance of a twice differentiated, smoothed white noise from order $n^{-1}h^{-d-4}$ to order $n^{-1}h^{-d-3}$. The small-radius assumption supplies uniform convexity of the pressure and uniform control of this transport geometry.

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